

Dressed Trajectory Fields: Generated Temporal Nonlocality in Conditioned Stochastic Dynamics

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Conditioning a nonlinear stochastic state process on an observation trajectory defines a path ensemble whose bare action is temporally local, while its inverse connected two-point function can acquire off-band structure through nonlinear fluctuation dressing. We develop a deterministic field-theoretic formulation of this generated temporal nonlocality. A general finite-horizon nonlinear, non-Gaussian state-space model is represented by a Euclidean trajectory action and source functional. Exact Schwinger–Dyson identities generate the hierarchy of trajectory observables, and a two-particle-irreducible (2PI) double Legendre transform promotes the trajectory mean and connected two-point function to independent fields. The resulting Dyson equation separates the block-tridiagonal bare Hessian from a fluctuation-generated self-energy. We identify a leading weak-coupling mechanism for nonzero off-band blocks, prove perturbative consistency of complete finite-order closures, establish exponential self-energy localization under correlation decay, and obtain controlled temporal-range truncation under explicit spectral and branch-stability conditions together with admissibility and block-banded solve bounds. Experiments on an exactly solvable nonlinear/non-Gaussian trajectory family with eight-dimensional states reproduce the predicted perturbative orders, exhibit exact-family self-energy localization and range decay, and confirm linear-in-horizon banded-solve scaling. Post-reference comparisons with mean-field Gaussian variational Bayes, a 12-sweep damped Gaussian expectation-propagation calculation, Laplace smoothing and an oracle iid sampler are kept separate from the field-theoretic construction.

Keywords: Schwinger–Dyson equations; 2PI effective action; nonlinear state-space models; trajectory inference; self-energy; nonlocality; deterministic inference.

1. Introduction

A state-space model is local by construction: the transition at time t couples x_{t-1} only to x_t , and the observation at time t couples the trajectory only to the local state x_t . This locality is often inherited by computational approximations. Gaussian smoothers use sparse Gaussian trajectory precision matrices, variational methods restrict the admissible path law, expectation propagation replaces non-Gaussian factors by tractable sites, and sampling methods approximate path expectations by finite empirical averages. These constructions are useful, but they obscure a structural distinction that becomes decisive outside the linear-Gaussian setting: locality of the *bare trajectory action* does not imply locality of the *inverse connected two-point function*.

The distinction is standard in effective-action methods. Schwinger–Dyson identities express exact integration-by-parts constraints on correlation functions [1, 2]; the two-particle-irreducible (2PI) effective action promotes the mean field and connected two-point function to independent variational objects [3, 4]. Field-theoretic and path-integral formulations of stochastic dynamics have a long history, including the Martin–Siggia–Rose construction, dynamical response functionals, and effective-action methods for Langevin and stochastic differential equations [5, 6, 7, 8, 9]. What is less commonly exploited in conditioned finite-horizon dynamical systems is that the same machinery can be applied directly to

the complete state trajectory while retaining a bilocal connected two-point field. The state dimension, the temporal horizon and the observation model enter only through the trajectory action. There is no requirement of a linear transition, Gaussian noise, additive observation error, stationarity or time homogeneity.

This viewpoint produces a useful inversion of the usual approximation logic. Instead of beginning from a filtering or smoothing algorithm and asking how to modify it for a nonlinear model, one begins from the exact conditioned path functional, derives the exact identities it must satisfy, and introduces approximation only at the level of the effective action. The resulting self-energy has an immediate dynamical meaning. The bare Hessian is block tridiagonal because the original action is nearest-neighbour in time, whereas the self-energy records interaction generated by integrating fluctuations. Consequently, nonzero off-band blocks of the inverse connected two-point function are not inserted by hand: they are generated by nonlinear fluctuation dressing in the conditioned path ensemble. Conditioning selects the ensemble studied here, but is not by itself the cause of the off-band structure; the same mechanism can occur in an unconditioned non-Gaussian Markov chain.

The paper makes five contributions. First, Section 2 gives a trajectory-field formulation for general positive smooth state-space densities, and Section 3 derives the exact Schwinger–Dyson hierarchy on the finite time grid. Second, Section 4 constructs the exact 2PI effective action and its mean and Dyson stationarity equations. Third, Section 5 proves structural results specific to conditioned trajectories: an exact inverse-two-point/self-energy identity, a leading weak-coupling mechanism for generated off-band structure, perturbative consistency of complete 2PI truncations, exponential self-energy localization under correlation decay, controlled range truncation, spectral admissibility and block-banded factorization/solve complexity. Fourth, Section 6 turns these equations into a deterministic dressed-trajectory procedure while keeping solver convergence logically separate from truncation accuracy. Fifth, Section 7 evaluates the theory on an exactly solvable multivariate nonlinear/non-Gaussian family. The experiments are designed to support theorem-predicted slopes and structural effects rather than to accumulate benchmark wins.

Only after the main development do we recast the same path law in Bayesian language. The post-reference appendix (Appendix) derives four competing solution families—mean-field Gaussian variational Bayes, Gaussian expectation propagation, trajectory Laplace approximation, and sampling-based smoothing—from the same target. This placement is intentional: the proposed method is defined by the trajectory action and its effective field theory, not by a probabilistic approximation family. The appendix is used solely to make comparisons on a common inferential object. Related mathematically controlled approximation programmes include state-evolution analyses of message-passing algorithms [10, 11] and graphical-model factorization and Gaussian Markov random-field methods [12, 13]; the present development is different in retaining a general finite trajectory and constructing its dressed two-point theory directly.

2. Conditioned trajectory action

Consider a finite horizon $t = 0, \dots, T$ with states $x_t \in \mathbb{R}^d$ and observations $y_t \in \mathbb{R}^{d_y}$. Let $p_0(x_0)$ be an initial density, $M_t(x_t | x_{t-1})$ a transition density and $\Lambda_t(y_t | x_t)$ an observation density. Throughout the main text the observed sequence $y_{0:T}$ is fixed and is therefore suppressed from some notation.

Definition 2.1 (Conditioned trajectory weight) *The unnormalized conditioned trajectory weight is*

$$\mathcal{P}_y[x_{0:T}] = p_0(x_0) \prod_{t=1}^T M_t(x_t | x_{t-1}) \prod_{t=0}^T \Lambda_t(y_t | x_t). \quad (2.1)$$

Its Euclidean trajectory action is

$$S_y[x] = -\log p_0(x_0) - \sum_{t=1}^T \log M_t(x_t | x_{t-1}) - \sum_{t=0}^T \log \Lambda_t(y_t | x_t). \quad (2.2)$$

It is convenient to collect all coordinates into $x = (x_0, \dots, x_T) \in \mathbb{R}^n$ with $n = (T+1)d$. The action can be written in nearest-neighbour form

$$S_y[x] = U_0(x_0) + \sum_{t=1}^T U_t(x_{t-1}, x_t), \quad (2.3)$$

where the observation contribution at time t is absorbed into one of the adjacent local potentials. This representation, rather than any Gaussian assumption, is the source of the bare temporal locality used below.

We impose the following finite-dimensional regularity conditions when an exact identity requires them.

Definition 2.2 (Admissible trajectory action) *An action $S_y : \mathbb{R}^n \rightarrow \mathbb{R}$ is called admissible if e^{-S_y} is integrable, S_y is continuously differentiable to the order used in the relevant statement, and every integration-by-parts boundary term appearing below vanishes. A convenient sufficient tail condition is that, outside a compact set,*

$$S_y(x) \geq c \|x\|^q - C \quad (2.4)$$

for constants $c > 0$, $q > 0$ and $C < \infty$, together with at most polynomial growth of the inserted test functions and of the derivatives of S_y used in the identity.

Whenever sources are introduced, we additionally assume that the corresponding partition function is finite and differentiable on an open source domain containing the physical point. For the linear source in (2.5), this means an open neighbourhood of $J = 0$ on which $Z_y[J] < \infty$ and the derivatives used below may be passed through the integral.

Introduce a linear source $J = (J_0, \dots, J_T)$ and define

$$Z_y[J] = \int_{\mathbb{R}^n} \exp[-S_y[x] + J^\top x] dx, \quad (2.5)$$

$$W_y[J] = \log Z_y[J]. \quad (2.6)$$

The connected trajectory mean and two-point function are

$$m_a[J] = \frac{\partial W_y}{\partial J_a}, \quad (2.7)$$

$$G_{ab}[J] = \frac{\partial^2 W_y}{\partial J_a \partial J_b}, \quad (2.8)$$

where indices a, b may denote either scalar coordinates or compound time-state indices. At $J = 0$ we write $m = m[0]$ and $G = G[0]$.

Remark 2.1 (Probability is not required for the calculus) *The normalized measure $Z_y[0]^{-1} e^{-S_y[x]} dx$ is useful notation, but the derivations in Sections 3–5 require only a finite-dimensional weighted integral and its Legendre transforms. A Bayesian interpretation of the same normalized object is postponed to the post-reference subsection [Bayesian recasting of the common trajectory problem](#).*

3. Exact Schwinger–Dyson hierarchy

The trajectory identities follow directly from integration by parts. They are exact before any closure is introduced.

Theorem 3.1 (Trajectory Schwinger–Dyson identity) *Let S_y be admissible and let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable with vanishing boundary terms. Then for every coordinate a ,*

$$\mathbb{E}[\partial_a F(X)] = \mathbb{E}[F(X) \partial_a S_y(X)], \quad (3.1)$$

where expectation is with respect to the normalized weight proportional to e^{-S_y} .

Proof Since the boundary term vanishes,

$$0 = \int_{\mathbb{R}^n} \partial_a \left(F(x) e^{-S_y(x)} \right) dx = \int_{\mathbb{R}^n} [\partial_a F(x) - F(x) \partial_a S_y(x)] e^{-S_y(x)} dx. \quad (3.2)$$

Division by $Z_y[0]$ gives (3.1). $\square$

Taking $F = 1$ and $F = x_b - m_b$ gives the first two exact identities

$$\mathbb{E}[\partial_a S_y(X)] = 0, \quad (3.3)$$

$$\mathbb{E}[(X_b - m_b) \partial_a S_y(X)] = \delta_{ab}. \quad (3.4)$$

For the local action (2.3), the trajectory score at an interior time is

$$\mathcal{S}_t(x) := \nabla_{x_t} S_y(x) = \nabla_2 U_t(x_{t-1}, x_t) + \nabla_1 U_{t+1}(x_t, x_{t+1}). \quad (3.5)$$

Thus (3.3) is local in its explicit arguments even though its expectation is taken under the full conditioned path law.

Monomial or tensor-valued choices of F generate an infinite hierarchy coupling moments of increasing order. Equivalently, source differentiation packages the hierarchy into a functional differential equation. For example, integration by parts in the sourced measure gives

$$\left[\partial_a S_y \left(\frac{\partial}{\partial J} \right) - J_a \right] Z_y[J] = 0, \quad (3.6)$$

whenever the operator substitution is well defined, with the corresponding connected version obtained after writing $Z_y = e^{W_y}$. The hierarchy is exact; approximation enters only when the higher connected functions are closed in terms of a finite collection of lower-order objects.

4. Two-particle-irreducible trajectory effective action

To retain the full connected two-point function as an independent object, introduce a symmetric quadratic source $K \in \text{Sym}(n)$. We assume that there is an open source domain $\mathcal{U}_{\text{src}} \subset \mathbb{R}^n \times \text{Sym}(n)$ containing $(0, 0)$ on which $Z_y[J, K]$ is finite and sufficiently differentiable for the double Legendre transform. Derivatives with respect to K are taken on the symmetric matrix space with Frobenius pairing

$$\langle A, B \rangle_F := \text{Tr}(AB), \quad D_K W[K](\Delta K) = \langle \nabla_K W, \Delta K \rangle_F, \quad (4.1)$$

which fixes the off-diagonal factor convention. With this convention,

$$Z_y[J, K] = \int_{\mathbb{R}^n} \exp \left[-S_y[x] + J^\top x + \frac{1}{2} x^\top K x \right] dx, \quad W_y[J, K] = \log Z_y[J, K]. \quad (4.2)$$

The derivatives are

$$m = \frac{\partial W_y}{\partial J}, \quad (4.3)$$

$$G + mm^\top = 2 \frac{\partial W_y}{\partial K}. \quad (4.4)$$

Definition 4.1 (2PI trajectory effective action) *On a branch for which (J, K) can be expressed as functions of (m, G) , we take the off-shell two-point field in the positive-definite cone $G \in \text{Sym}_{++}(n)$. This is the natural Legendre branch on which the matrix logarithm in (4.10) is real and well defined. Define the double Legendre transform*

$$\Gamma_{2,y}[m, G] = -W_y[J, K] + J^\top m + \frac{1}{2} \text{Tr} \left[K(G + mm^\top) \right]. \quad (4.5)$$

Differentiation of (4.5) gives

$$\nabla_m \Gamma_{2,y} = J + Km, \quad (4.6)$$

$$\nabla_G \Gamma_{2,y} = \frac{1}{2} K. \quad (4.7)$$

Hence the physical point $(m_\star, G_\star)$ at $J = K = 0$ is stationary:

$$\nabla_m \Gamma_{2,y}[m_\star, G_\star] = 0, \quad \nabla_G \Gamma_{2,y}[m_\star, G_\star] = 0. \quad (4.8)$$

Let

$$H_y[m] := \nabla^2 S_y[m] \quad (4.9)$$

be the bare trajectory Hessian and shift $x = m + \phi$. The standard finite-dimensional 2PI decomposition [3] can be written in the present sign convention as

$$\Gamma_{2,y}[m, G] = S_y[m] + \frac{1}{2} \text{Tr} \log G^{-1} + \frac{1}{2} \text{Tr} [H_y[m]G - I] + \Phi_y[m, G] + \mathcal{C}_y, \quad (4.10)$$

where $\Phi_y[m, G]$ is defined *exactly* by (4.10) as the interaction remainder of the double Legendre transform, and $\mathcal{C}_y$ is independent of (m, G) . No convergence of a diagrammatic series is assumed in this identity. If

the action is embedded in a regular perturbative family, the formal or analytic expansion coefficients of Φ_y are represented by 2PI skeleton diagrams built from the shifted interaction vertices and the dressed propagator G . Define the exact self-energy

$$\Sigma_y[m, G] := -2\nabla_G \Phi_y[m, G]. \quad (4.11)$$

Stationarity of (4.10) yields the exact Dyson equation

$$G^{-1} = H_y[m] - \Sigma_y[m, G], \quad (4.12)$$

and the exact mean equation

$$0 = \nabla S_y[m] + \frac{1}{2} \nabla_m \text{Tr}[H_y[m]G] + \nabla_m \Phi_y[m, G]. \quad (4.13)$$

The distinction between (4.12) and an ordinary local Gaussian approximation is central. The matrix $H_y[m]$ inherits the nearest-neighbour structure of the bare action, whereas Σ_y contains fluctuation-induced interaction generated by the full path ensemble. Throughout the paper, the phrase *dressed precision* is used only as shorthand for the inverse connected two-point function G^{-1} . Outside the Gaussian case, G^{-1} is not a precision matrix in the Gaussian graphical-model sense and does not encode conditional independence. The next section makes the inverse-two-point statement precise.

5. Structural theorems for dressed trajectories

This section records properties that are intrinsic to the trajectory-field construction. They do not compare the method with an external inference algorithm; those comparisons are deferred to the post-reference appendix, [Appendix](#). Because the observation trajectory y is fixed throughout, this section suppresses its subscript when no ambiguity can arise: $H[m] := H_y[m]$ and $\Sigma[m, G] := \Sigma_y[m, G]$. Exact physical quantities carry a star.

5.1. Bare locality and generated temporal nonlocality

Lemma 5.1 (Block locality of the bare Hessian) *For an action of the nearest-neighbour form (2.3), the block Hessian satisfies*

$$H_{y,ts}[m] = 0, \quad |t - s| > 1. \quad (5.1)$$

Proof Every term in (2.3) depends either on a single state or on one adjacent pair (x_{t-1}, x_t) . A mixed second derivative with respect to x_t and x_s therefore vanishes whenever the two time indices do not occur in a common local potential, which is exactly the condition $|t - s| > 1$. $\square$

Theorem 5.1 (Exact generated-nonlocality identity) *At the exact physical 2PI stationary point, every off-band block of the inverse connected two-point function is generated entirely by the self-energy:*

$$(G_\star^{-1})_{ts} = -(\Sigma_\star)_{ts}, \quad |t - s| > 1. \quad (5.2)$$

Consequently, if $(\Sigma_\star)_{ts} \neq 0$ for some $|t - s| > 1$, then $G_\star^{-1}$ is not block banded despite the nearest-neighbour factorization of the conditioned path density.

Proof Take the (t, s) block of the exact Dyson equation (4.12). By Lemma 5.1, $H_{y,ts} = 0$ for $|t - s| > 1$, giving (5.2) immediately. $\square$

Remark 5.1 (Inverse covariance is not a Markov graph outside the Gaussian case) *Even a Gaussian Markov chain has a dense covariance G . Theorem 5.1 concerns the inverse connected two-point function. In a purely Gaussian local chain, G^{-1} is the Gaussian precision matrix and is block tridiagonal. For a non-Gaussian chain, however, a dense G^{-1} does not imply loss of Markov conditional independence: the density still factorizes into the nearest-neighbour potentials in (2.3) and therefore remains a chain-structured Markov random field. The off-band blocks quantify nonlocality of the connected two-point response, not a change in the graphical-model factorization.*

Remark 5.2 (Coordinate dependence of the dressed two-point nonlocality) *The locality of the path-density factorization is preserved by invertible transformations applied separately within each time slice, but the covariance and its inverse are not invariant under nonlinear reparameterization. Hence the off-band structure of G^{-1} and the associated self-energy are properties of the chosen state-field coordinates. This is visible in the manufactured family of Section 7: in the Gaussian u coordinates the inverse connected two-point function is block banded, whereas the nonlinear coordinates $x = \varphi_\varepsilon(u)$ produce an off-band tail. The structural statements in this paper therefore refer to the physically chosen trajectory field x , not to a nonlinear-coordinate invariant notion of memory.*

Proposition 5.1 (Leading cubic mechanism for off-band self-energy) *Let*

$$S_\varepsilon(x) = S_0(x) + \varepsilon V(x) + O(\varepsilon^2), \quad S_0(x) = \frac{1}{2}(x - m_0)^\top H_0(x - m_0), \quad (5.3)$$

with $H_0 \succ 0$, $G_0 = H_0^{-1}$, and a nearest-neighbour interaction V . Assume moreover that S_ε retains the nearest-neighbour form (2.3) for all sufficiently small $|\varepsilon|$, so in particular H_0 is block tridiagonal and every perturbative coefficient of the bare action remains temporally local. Write

$$T_{abc} := \partial_a \partial_b \partial_c V(m_0). \quad (5.4)$$

In the sign convention of (4.12), the two-cubic-vertex 2PI skeleton gives one contribution to the order- ε^2 self-energy coefficient, evaluated at the reference pair (m_0, G_0) :

$$(\Sigma_2^{\text{cc}})_{ad} = \frac{1}{2} \sum_{b,c,e,f} T_{abc} T_{def} (G_0)_{be} (G_0)_{cf} \quad (5.5)$$

to the coefficient in the fixed-reference expansion

$$\Sigma[m_0, G_0; \varepsilon] = \varepsilon \Sigma_1[m_0, G_0] + \varepsilon^2 \Sigma_2[m_0, G_0] + O(\varepsilon^3). \quad (5.6)$$

In particular, for external indices $a = (t, i)$ and $d = (t + 2, \ell)$, any nonzero mixed local cubic vertices on the adjacent pairs $(t, t + 1)$ and $(t + 1, t + 2)$ give a generally nonzero lag-two contribution whenever the contraction in (5.5) is nonzero. Thus, within the class of local interactions with nonvanishing cubic vertices, off-band inverse-two-point structure is generically present already at second weak-coupling order.

Proof The cubic part of the shifted interaction is $(\varepsilon/3!)T_{abc}\phi_a\phi_b\phi_c$. The lowest 2PI vacuum diagram containing two such vertices is the sunset term

$$\Phi_{\text{cc}}[G] = -\frac{\varepsilon^2}{12}T_{abc}T_{def}G_{ad}G_{be}G_{cf}, \quad (5.7)$$

with repeated indices summed. Applying $\Sigma = -2\nabla_G\Phi$ yields (5.5). Locality restricts the indices within each cubic vertex but does not force the two dressed propagators joining the vertices to vanish. The lag-two statement follows by choosing the two local vertex supports on consecutive temporal bonds. Nonvanishing of the displayed contraction is generic; its failure requires a zero vertex, a zero propagator contraction, or an exact algebraic cancellation. $\square$

5.2. Controlled weak-coupling truncation

Introduce a genuine weak-coupling parameter ε and suppose the exact 2PI functional has a regular branch

$$\Gamma_{2,\varepsilon}(z), \quad z := (m, G), \quad (5.8)$$

near $\varepsilon = 0$. Let

$$F(z, \varepsilon) := \nabla_z \Gamma_{2,\varepsilon}(z) \quad (5.9)$$

be the exact stationarity map. Square-bracket superscripts $[p]$ will denote truncation in powers of ε through perturbative order p ; angle-bracket superscripts $\langle L \rangle$ below will denote a fixed finite retained 2PI skeleton family indexed collectively by L , which is a logically different notion.

Lemma 5.2 (Complete 2PI functional truncation implies complete stationarity truncation) *Suppose on a neighbourhood $\mathcal{U}_z$ of z_0 the exact 2PI functional admits*

$$\Gamma_{2,\varepsilon}(z) = \sum_{r=0}^p \varepsilon^r \Gamma_{2,r}(z) + R_{p+1}(z, \varepsilon), \quad (5.10)$$

where $\Gamma_{2,r}$ is C^2 in z and, uniformly for $z \in \mathcal{U}_z$,

$$\|\nabla_z R_{p+1}(z, \varepsilon)\| + \|D_z \nabla_z R_{p+1}(z, \varepsilon)\| \leq C|\varepsilon|^{p+1}. \quad (5.11)$$

Define the complete perturbative truncation

$$\Gamma_{2,\varepsilon}^{[p]}(z) := \sum_{r=0}^p \varepsilon^r \Gamma_{2,r}(z), \quad F^{[p]} := \nabla_z \Gamma_{2,\varepsilon}^{[p]}. \quad (5.12)$$

Then

$$F^{[p]} - F = O(\varepsilon^{p+1}), \quad D_z F^{[p]} - D_z F = O(\varepsilon^{p+1}) \quad (5.13)$$

uniformly on $\mathcal{U}_z$.

Proof Differentiate (5.10) once and twice with respect to z and use (5.11). $\square$

Theorem 5.2 (Perturbative consistency of a complete p th-order closure) *Assume that:*

1. F and $F^{[p]}$ are C^1 in z and analytic (or C^{p+1}) in ε in a neighbourhood of $(z_0, 0)$;
2. $F(z_0, 0) = 0$ and the Jacobian $D_z F(z_0, 0)$ is nonsingular;
3. the approximate stationarity map satisfies, uniformly in that neighbourhood,

$$F^{[p]}(z, \varepsilon) - F(z, \varepsilon) = O(\varepsilon^{p+1}), \quad D_z F^{[p]}(z, \varepsilon) - D_z F(z, \varepsilon) = O(\varepsilon^{p+1}). \quad (5.14)$$

Then for sufficiently small $|\varepsilon|$ there exist unique local stationary branches $z_\star(\varepsilon)$ and $z^{[p]}(\varepsilon)$ with $z_\star(0) = z^{[p]}(0) = z_0$, and

$$\|z^{[p]}(\varepsilon) - z_\star(\varepsilon)\| \leq C|\varepsilon|^{p+1}. \quad (5.15)$$

Equivalently, the Taylor coefficients of m and G agree with the exact coefficients through order p .

Proof The implicit-function theorem applied to F gives a unique exact branch $z_\star(\varepsilon)$ for sufficiently small $|\varepsilon|$. Assumption (5.14) and nonsingularity of $D_z F(z_0, 0)$ imply that $D_z F^{[p]}$ remains nonsingular in a possibly smaller neighbourhood, yielding a unique approximate branch. Evaluate the approximate stationarity map at the exact solution:

$$F^{[p]}(z_\star(\varepsilon), \varepsilon) = F(z_\star(\varepsilon), \varepsilon) + O(\varepsilon^{p+1}) = O(\varepsilon^{p+1}). \quad (5.16)$$

Set $\Delta z := z^{[p]}(\varepsilon) - z_\star(\varepsilon)$. The fundamental theorem of calculus along the line segment joining the two solutions gives

$$\begin{aligned} & F^{[p]}(z^{[p]}(\varepsilon), \varepsilon) - F^{[p]}(z_\star(\varepsilon), \varepsilon) \\ &= \left[\int_0^1 D_z F^{[p]}(z_\star(\varepsilon) + s\Delta z, \varepsilon) \, ds \right] \Delta z. \end{aligned} \quad (5.17)$$

For sufficiently small $|\varepsilon|$ and a sufficiently small common branch neighbourhood, the integral Jacobian is uniformly close to the nonsingular reference matrix $D_z F(z_0, 0)$ and therefore has a uniformly bounded inverse. Since the first term on the left vanishes and $F^{[p]}(z_\star(\varepsilon), \varepsilon) = O(\varepsilon^{p+1})$, multiplication by that inverse yields (5.15). Equality of Taylor coefficients through order p follows by differentiating the two stationary branches at $\varepsilon = 0$. $\square$

Corollary 5.1 (Complete perturbative 2PI closure) *Under the hypotheses of Lemma 5.2 and Theorem 5.2, retaining the complete perturbative coefficients of the exact 2PI functional through order p produces a stationary branch satisfying (5.15). In particular, retaining a set of skeleton diagrams is controlled by Theorem 5.2 only when that set generates every coefficient of $\Gamma_{2,\varepsilon}$ required through the stated power of ε .*

Remark 5.3 (What “order” means) *Theorem 5.2 does not say that an arbitrary diagram truncation is controlled. The hypothesis is completeness of the stationarity equations through the stated power of a genuine expansion parameter. A finite retained skeleton family indexed collectively by L need not coincide with perturbative order p . A diagram family selected by loop or topology criteria can mix several powers of ε , while a self-consistent solution can resum higher powers. Perturbative control therefore requires retaining every contribution that enters through order p ; self-consistency may resum additional terms, but it does not by itself promote an incomplete diagram selection to a controlled one.*

5.3. Self-energy localization and range truncation

The trajectory field contains a second approximation axis besides the choice of retained finite skeleton family: the temporal range retained in the self-energy. The next result provides a sufficient condition under which that truncation is controlled.

Theorem 5.3 (Exponential localization for a finite retained skeleton family) *Consider a fixed finite retained family of 2PI skeletons, indexed collectively by a truncation level L . Assume that the local vertex tensors entering this retained family, together with their temporal-diameter bounds, are uniformly bounded independently of the horizon T , and that there exist constants $C_G < \infty$ and $\alpha > 0$, also independent of T , such that the dressed propagator satisfies*

$$\|G_{ts}\| \leq C_G e^{-\alpha|t-s|}. \quad (5.18)$$

Then for every $0 < \alpha' < \alpha$ there exists $C_{L,\alpha'} < \infty$, independent of both T and the external times, such that

$$\left\| \Sigma_{ts}^{(L)} \right\| \leq C_{L,\alpha'} e^{-\alpha'|t-s|}. \quad (5.19)$$

Proof Differentiating the fixed finite retained family of 2PI skeletons gives a finite family of connected self-energy diagrams. Consider one such diagram and collapse the bounded temporal support of each local vertex to one integer time label; the resulting finite graph has two external vertices carrying t and s . Choose a spanning tree $\mathcal{T}$ of this graph and let $\mathcal{P} \subset \mathcal{T}$ be the unique tree path joining the two external vertices. Put $\delta := \alpha - \alpha' > 0$. Ignoring non-tree propagator exponentials only enlarges the bound, and for the tree edges

$$\begin{aligned} \prod_{(ij) \in \mathcal{T}} e^{-\alpha|r_i-r_j|} &= \left[\prod_{(ij) \in \mathcal{P}} e^{-\alpha'|r_i-r_j|} \right] \left[\prod_{(ij) \in \mathcal{P}} e^{-\delta|r_i-r_j|} \right] \left[\prod_{(ij) \in \mathcal{T} \setminus \mathcal{P}} e^{-\alpha|r_i-r_j|} \right] \\ &\leq e^{-\alpha'|t-s|} \prod_{(ij) \in \mathcal{T}} e^{-\delta|r_i-r_j|}, \end{aligned} \quad (5.20)$$

where the triangle inequality was used on $\mathcal{P}$ and $e^{-\alpha r} \leq e^{-\delta r}$. Let $k_\delta(r) := e^{-\delta|r|}$. Summing the internal time labels by successively removing leaves of the tree contributes at most one factor

$$\|k_\delta\|_{\ell^1(\mathbb{Z})} = 1 + 2 \sum_{r \geq 1} e^{-\delta r} < \infty \quad (5.21)$$

per eliminated internal vertex. Thus every diagram is bounded by a constant, depending only on its finite combinatorics, the vertex bounds, C_G , and δ , times $e^{-\alpha'|t-s|}$. The bounded temporal diameter of a local vertex changes only that constant. Summing the finite diagram family yields (5.19) with a constant independent of the horizon T . $\square$

For an integer range $R \geq 0$, define the range-projected self-energy

$$(\Sigma^{(L,R)})_{ts} := \begin{cases} (\Sigma^{(L)})_{ts}, & |t-s| \leq R, \\ 0, & |t-s| > R. \end{cases} \quad (5.22)$$

We emphasize that (5.22) is used as a computational projection of a retained finite- L self-energy. If a variationally closed range-restricted functional is desired, the same sparsity constraint can instead be imposed at the level of the admissible 2PI family.

Corollary 5.2 (Covariance-update error under range projection) *Let*

$$A := H - \Sigma^{\langle L \rangle}, \quad A_R := H - \Sigma^{\langle L, R \rangle}, \quad (5.23)$$

and suppose $A \succeq \mu I$ and $A_R \succeq \mu I$ for some $\mu > 0$. If Theorem 5.3 holds, then for every $0 < \alpha' < \alpha$ there exists $C_{L, \alpha'}$ independent of both R and T such that

$$\|A^{-1} - A_R^{-1}\|_2 \leq \frac{C_{L, \alpha'}}{\mu^2} e^{-\alpha' R}. \quad (5.24)$$

For the full $n \times n$ trajectory matrix, $n = (T + 1)d$, the corresponding Frobenius estimate is

$$\|A^{-1} - A_R^{-1}\|_F \leq \sqrt{(T + 1)d} \|A^{-1} - A_R^{-1}\|_2, \quad (5.25)$$

so an explicitly uniform Frobenius bound requires additional structure if T is allowed to grow. This corollary is a fixed-input update bound: H and both self-energies are evaluated at the same pair (m, G) . It does not by itself compare the stationary points obtained after the two closure systems are separately re-solved.

Proof Put $E_R := \Sigma^{\langle L \rangle} - \Sigma^{\langle L, R \rangle}$. For a block matrix the Schur bound gives

$$\|E_R\|_2 \leq \sqrt{\left(\sup_t \sum_s \|(E_R)_{ts}\|_2 \right) \left(\sup_s \sum_t \|(E_R)_{ts}\|_2 \right)}. \quad (5.26)$$

Here E_R is symmetric, so the two factors coincide. Using (5.19), each block row tail is bounded by

$$\sup_t \sum_{|t-s| > R} \left\| \Sigma_{ts}^{\langle L \rangle} \right\|_2 \leq 2C_{L, \alpha'} \sum_{r > R} e^{-\alpha' r} \leq \tilde{C}_{L, \alpha'} e^{-\alpha' R}, \quad (5.27)$$

with a constant independent of R and T . Absorbing the geometric-series factor into the constant therefore yields

$$\left\| \Sigma^{\langle L \rangle} - \Sigma^{\langle L, R \rangle} \right\|_2 \leq \tilde{C}_{L, \alpha'} e^{-\alpha' R}. \quad (5.28)$$

The resolvent identity gives

$$A_R^{-1} - A^{-1} = A_R^{-1}(A - A_R)A^{-1}. \quad (5.29)$$

The spectral lower bounds imply $\|A^{-1}\|_2, \|A_R^{-1}\|_2 \leq \mu^{-1}$, proving (5.24). $\square$

Theorem 5.4 (Stability of the re-solved range-truncated branch) *Let $z = (m, G)$ and let*

$$F^{\langle L \rangle}(z) = 0 \quad (5.30)$$

denote the finite- L closure equations, with a solution $z_\star^{\langle L \rangle}$. Let

$$F^{\langle L, R \rangle}(z) = 0 \quad (5.31)$$

denote the corresponding range-projected closure equations. Assume that $F^{\langle L \rangle}$ is C^1 on a neighbourhood $\mathcal{U}_z$ of $z_\star^{\langle L \rangle}$, that

$$D_z F^{\langle L \rangle}(z_\star^{\langle L \rangle}) \quad (5.32)$$

is nonsingular, and that for some independent rate $\gamma > 0$ and all sufficiently large R ,

$$\sup_{z \in \mathcal{U}_z} \left[\left\| F^{\langle L, R \rangle}(z) - F^{\langle L \rangle}(z) \right\| + \left\| D_z F^{\langle L, R \rangle}(z) - D_z F^{\langle L \rangle}(z) \right\| \right] \leq C e^{-\gamma R}. \quad (5.33)$$

Then, for all sufficiently large R , there is a unique local range-truncated solution $z_\star^{\langle L, R \rangle}$ near $z_\star^{\langle L \rangle}$ and a constant C' independent of R such that

$$\left\| z_\star^{\langle L, R \rangle} - z_\star^{\langle L \rangle} \right\| \leq C' e^{-\gamma R}. \quad (5.34)$$

Proof Set

$$J_\star := D_z F^{\langle L \rangle}(z_\star^{\langle L \rangle}), \quad (5.35)$$

which is invertible by assumption, and define on a closed ball $B_\rho(z_\star^{\langle L \rangle}) \subset \mathcal{U}_z$

$$T_R(z) := z - J_\star^{-1} F^{\langle L, R \rangle}(z). \quad (5.36)$$

By continuity of $D_z F^{\langle L \rangle}$, choose $\rho > 0$ so that

$$\sup_{z \in B_\rho} \left\| I - J_\star^{-1} D_z F^{\langle L \rangle}(z) \right\| \leq q_0 < 1. \quad (5.37)$$

The Jacobian estimate in (5.33) then gives, for all sufficiently large R ,

$$\sup_{z \in B_\rho} \|DT_R(z)\| \leq q < 1 \quad (5.38)$$

for some q independent of R . Moreover,

$$\begin{aligned} \left\| T_R(z_\star^{\langle L \rangle}) - z_\star^{\langle L \rangle} \right\| &= \left\| J_\star^{-1} F^{\langle L, R \rangle}(z_\star^{\langle L \rangle}) \right\| \\ &\leq \|J_\star^{-1}\| C e^{-\gamma R}. \end{aligned} \quad (5.39)$$

For sufficiently large R this is at most $(1 - q)\rho$, and therefore, for every $z \in B_\rho$,

$$\left\| T_R(z) - z_\star^{\langle L \rangle} \right\| \leq q\rho + (1 - q)\rho = \rho. \quad (5.40)$$

Hence T_R maps the closed ball into itself and is a contraction. Banach's fixed-point theorem gives a unique fixed point $z_\star^{\langle L,R \rangle}$ in that ball, equivalently a unique nearby zero of $F^{\langle L,R \rangle}$. Finally,

$$\left\| z_\star^{\langle L,R \rangle} - z_\star^{\langle L \rangle} \right\| \leq q \left\| z_\star^{\langle L,R \rangle} - z_\star^{\langle L \rangle} \right\| + \|J_\star^{-1}\| C e^{-\gamma R}. \quad (5.41)$$

so

$$\left\| z_\star^{\langle L,R \rangle} - z_\star^{\langle L \rangle} \right\| \leq \frac{\|J_\star^{-1}\| C}{1-q} e^{-\gamma R}, \quad (5.42)$$

which proves (5.34). $\square$

Remark 5.4 (Additional stability hypothesis) *The Jacobian estimate in (5.33) is an additional branch-stability hypothesis. Theorem 5.3 alone establishes localization of the retained self-energy; it does not by itself establish localization of all derivatives of the closure map with respect to (m, G) .*

The use of exponential localization is closely related to classical decay results for inverses of banded positive-definite matrices [14]. Here the direction is different: the propagator decay is an assumption controlling the spatial reach of finite-order self-energy diagrams, after which the self-energy tail itself can be truncated.

5.4. Spectral admissibility and computational scaling

Theorem 5.5 (Pointwise spectral admissibility) *At any pair (m, G) for which the bare Hessian and retained self-energy satisfy*

$$H[m] \succeq hI, \quad \left\| \Sigma^{\langle L,R \rangle}[m, G] \right\|_2 \leq \sigma < h, \quad (5.43)$$

the dressed inverse-two-point operator obeys

$$H[m] - \Sigma^{\langle L,R \rangle}[m, G] \succeq (h - \sigma)I, \quad (5.44)$$

and the covariance update

$$G_{\text{new}} = \left(H[m] - \Sigma^{\langle L,R \rangle}[m, G] \right)^{-1} \quad (5.45)$$

is positive definite with

$$\|G_{\text{new}}\|_2 \leq (h - \sigma)^{-1}. \quad (5.46)$$

If the two inequalities in (5.43) hold uniformly on an iteration domain $\mathcal{D}$, the same admissibility conclusion holds at every update whose inputs lie in $\mathcal{D}$.

Proof For every unit vector v ,

$$v^\top (H - \Sigma^{\langle L,R \rangle}) v \geq h - |v^\top \Sigma^{\langle L,R \rangle} v| \geq h - \left\| \Sigma^{\langle L,R \rangle} \right\|_2 \geq h - \sigma > 0. \quad (5.47)$$

The inverse bound follows from the reciprocal of the smallest eigenvalue. $\square$

Theorem 5.6 (Block-banded factorization and solve complexity) *If $H - \Sigma^{\langle L, R \rangle}$ is symmetric positive definite and has block half-bandwidth $R_{\text{eff}} = \max\{1, R\}$ with $d \times d$ blocks, then its block-banded Cholesky factorization and a fixed number of triangular linear solves have arithmetic complexity*

$$O(TR_{\text{eff}}^2 d^3) \quad (5.48)$$

and storage complexity

$$O(TR_{\text{eff}} d^2). \quad (5.49)$$

For fixed R and d , these factorization and solve costs are therefore $O(T)$. The statement does not include explicit formation of the dense inverse covariance G , which would require $O(T^2 d^2)$ storage in general.

Proof At each of the $T + 1$ block rows, block elimination interacts with $O(R_{\text{eff}})$ neighbouring blocks and performs $O(R_{\text{eff}}^2)$ dense $d \times d$ block products or solves. Each such operation costs $O(d^3)$. Storage retains $O(R_{\text{eff}})$ dense blocks per block row, each of size d^2 . $\square$

Combining Theorems 5.4–5.6, under their respective hypotheses, a target range-truncation branch error relative to the finite- L branch δ can be reached with $R = O(\log(1/\delta))$ whenever the hypotheses and constants remain uniform, giving the linear-algebra scaling

$$O\left(T d^3 \log^2 \frac{1}{\delta}\right). \quad (5.50)$$

This bound concerns the structured factorization and solves. If a chosen skeleton closure requires only covariance blocks in the elimination sparsity pattern or a fixed-width neighbourhood, those selected inverse blocks can be recovered by Takahashi/Erisman–Tinney-type selected-inversion recursions at the same order as the banded factorization [15]. If a closure requires the complete dense covariance, no end-to-end $O(T)$ claim is made. The cost of assembling the retained local vertices and skeleton contractions must also be added for the particular truncation.

6. Deterministic dressed-trajectory closure

The exact equations (4.12)–(4.13) are not generally closed in finite form because the exact interaction remainder Φ_y is not generally available in closed form. A practical method therefore chooses two independent approximation controls: a finite retained family of 2PI skeletons indexed collectively by L and, when justified by Theorem 5.3, a temporal self-energy range R . Perturbative order p is used separately only when a genuine weak-coupling expansion is invoked.

Write

$$\Phi_y[m, G] \approx \Phi_y^{\langle L \rangle}[m, G], \quad \Sigma_y^{\langle L \rangle}[m, G] := -2\nabla_G \Phi_y^{\langle L \rangle}[m, G]. \quad (6.1)$$

The corresponding range-controlled *closure equations* are

$$0 = \nabla S_y[m] + \frac{1}{2} \nabla_m \text{Tr}[H_y[m]G] + \nabla_m \Phi_y^{\langle L \rangle}[m, G], \quad (6.2)$$

$$G^{-1} = H_y[m] - \Sigma_y^{\langle L, R \rangle}[m, G]. \quad (6.3)$$

A generic deterministic solve alternates the following steps: evaluate the local action derivatives at the current mean; assemble the retained 2PI skeleton terms and their self-energy; apply the range

projection if used; factor the block-banded dressed inverse two-point operator; obtain the linear solves and any selected covariance blocks required by the retained contractions; update the mean equation by a damped Newton or quasi-Newton step; and continue until both closure residuals are small. The notation

$$\tilde{G}^{(k+1)} = \left(H_y[m^{(k)}] - \Sigma_y^{(L,R)}[m^{(k)}, G^{(k)}] \right)^{-1} \quad (6.4)$$

is conceptual: an implementation need not, and for linear scaling must not, materialize the full dense inverse. When only selected blocks are required, they are recovered from the banded factors. If a dense G is required by the chosen skeleton contractions, the $O(T)$ linear-algebra statement no longer applies end to end. A damped update may be written abstractly as

$$G^{(k+1)} = (1 - \eta_G)G^{(k)} + \eta_G \tilde{G}^{(k+1)}, \quad 0 < \eta_G \leq 1, \quad (6.5)$$

with the understanding that the stored representation may consist only of selected blocks.

Because the projection (5.22) is applied after differentiation, it need not be the gradient of any range-restricted 2PI functional. The numerical residuals are therefore defined directly from the closure equations,

$$r_m := \left\| \nabla S_y[m] + \frac{1}{2} \nabla_m \text{Tr}[H_y[m]G] + \nabla_m \Phi_y^{(L)}[m, G] \right\|, \quad (6.6)$$

$$r_G := \left\| G^{-1} - H_y[m] + \Sigma_y^{(L,R)}[m, G] \right\|_F. \quad (6.7)$$

They measure numerical satisfaction of the selected closure equations, while $\lambda_{\min}(G)$ or $\lambda_{\min}(G^{-1})$ monitors admissibility. Only if the range constraint is imposed at the functional level may these equations also be called stationarity conditions of a range-restricted 2PI functional.

Remark 6.1 (Solver convergence versus truncation accuracy) *Small values of (6.6)–(6.7) certify only numerical solution of the selected closure equations. Approximation error requires separate control. Theorem 5.2 together with Corollary 5.1 gives perturbative stationary-branch accuracy; Corollary 5.2 controls the fixed-input covariance-update error caused by range projection; and Theorem 5.4 controls the re-solved range-truncated branch under its additional stability hypotheses.*

The numerical study below isolates these two issues deliberately. The manufactured test family has exact conditioned moments, so perturbative and range errors can be measured without contamination by an approximate ground truth. For the perturbative plots, the order- p dressed Taylor polynomial is evaluated directly from the complete weak-coupling series through order p rather than through an iterative numerical skeleton solver. Proposition A.3 proves that these coefficients are exactly the coefficients of the complete perturbative 2PI stationary branch through the same order. The experiment therefore tests the perturbative 2PI prediction without conflating it with optimization error of a particular fixed-point implementation.

7. Numerical experiments

The experiments are designed to test the structural statements of Section 5, not to establish an incremental ranking on unrelated benchmark data. Unless otherwise stated, trajectory-statistic curves are means

over ten independently generated conditioned trajectories, with shaded regions showing one empirical standard deviation. Figure 6 uses timing repetitions, and Figures 10–11 use independent Monte Carlo repetitions as specified below. No standard-error bands are used.

We use one exactly solvable multivariate family throughout. Let $z_t \in \mathbb{R}^d$ and $w_t \in \mathbb{R}^d$ obey the latent linear-Gaussian chain

$$z_0 \sim \mathcal{N}(0, P_0), \quad (7.1)$$

$$z_t = Az_{t-1} + \xi_t, \quad \xi_t \sim \mathcal{N}(0, Q), \quad (7.2)$$

$$w_t = Cz_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, R_y). \quad (7.3)$$

Set $u_t = Uz_t$ and define the physical state coordinate componentwise by

$$x_t = \varphi_\varepsilon(u_t), \quad (7.4)$$

$$\varphi_\varepsilon(u) = u + a_\varepsilon \sin u, \quad a_\varepsilon := 1 - e^{-\varepsilon}, \quad (7.5)$$

while the reported observation coordinate remains

$$y_t = \frac{\sinh(\beta w_t)}{\beta}. \quad (7.6)$$

For the analytic weak-coupling family, ε is the independent deformation parameter. For numerical reporting only, we introduce a separate nonnegative parameter λ and set $\varepsilon = \lambda^2$. Because $|a_\varepsilon| < 1$ for the open interval $\varepsilon \in (-\log 2, \infty)$,

$$\frac{d\varphi_\varepsilon}{du} = 1 + a_\varepsilon \cos u \geq 1 - |a_\varepsilon| > 0, \quad (7.7)$$

so the state map is a smooth global diffeomorphism throughout that open interval. The numerical experiments subsequently restrict to $\varepsilon = \lambda^2 \geq 0$. Moreover

$$|\varphi_\varepsilon(u) - u| \leq |a_\varepsilon|, \quad (7.8)$$

so the transformation is asymptotically linear and differs from the Gaussian coordinate by a bounded amount. Consequently, if $u \mid y$ is Gaussian, the pushed-forward trajectory $x \mid y$ retains Gaussian-strength tails. In particular, for the finite trajectory dimension $n = (T+1)d$, there is an open neighbourhood of $(J, K) = (0, 0)$ on which

$$\mathbb{E} \exp \left(J^\top X + \frac{1}{2} X^\top K X \right) < \infty. \quad (7.9)$$

Thus the manufactured family satisfies the source-domain hypothesis of Section 4. It also supplies a locally invertible source-to-moment map at the physical point. Indeed, the transformed trajectory law has a smooth strictly positive density on all of $\mathbb{R}^n$. The Hessian of W with respect to the linear and symmetric-quadratic source coordinates is the covariance matrix of the sufficient statistics

$$\left(X, \frac{1}{2} X X^\top \right). \quad (7.10)$$

If a nonzero linear-plus-quadratic combination of these statistics had zero variance, the corresponding nonconstant polynomial would be almost surely constant; full support of the density would then force

that polynomial to be constant on all of $\mathbb{R}^n$, a contradiction. Hence this covariance is positive definite on the linear-plus-symmetric-quadratic source space. The change of variables from $(m, \mathbb{E}[XX^\top])$ to (m, G) , with $G = \mathbb{E}[XX^\top] - mm^\top$, is smooth and invertible. The inverse-function theorem therefore gives a local source-to- (m, G) inverse, so the exact $W[J, K]$ and local double Legendre branch used in Section 4 exist for this family.

For $\varepsilon \neq 0$ the induced transition density in the physical x coordinates is nonlinear and non-Gaussian. The state-coordinate Jacobian depends on x and enters the trajectory action. The observation transformation plays a different role: for fixed observed y , $w = \text{asinh}(\beta y)/\beta$ is known and its Jacobian is independent of the latent trajectory, so it contributes only an x -independent normalization. Writing $\psi_\varepsilon := \varphi_\varepsilon^{-1}$,

$$z_t = U^{-1} \psi_\varepsilon(x_t), \quad w_t = \frac{\text{asinh}(\beta y_t)}{\beta}. \quad (7.11)$$

The latent Gaussian conditional law is used only to construct exact reference quantities.

The conditional law of the stacked u trajectory is Gaussian,

$$u \mid y \sim \mathcal{N}(\mu, \mathcal{G}). \quad (7.12)$$

Let $v_i := \mathcal{G}_{ii}$ and define

$$b_i := e^{-v_i/2} \sin \mu_i, \quad c_i := e^{-v_i/2} \cos \mu_i, \quad (7.13)$$

$$\Xi_{ij} := \frac{1}{2} \left[e^{-\frac{1}{2}(v_i+v_j-2\mathcal{G}_{ij})} \cos(\mu_i - \mu_j) - e^{-\frac{1}{2}(v_i+v_j+2\mathcal{G}_{ij})} \cos(\mu_i + \mu_j) \right]. \quad (7.14)$$

Then the exact transformed mean is

$$(m_\star)_i = \mu_i + a_\varepsilon b_i, \quad (7.15)$$

and the exact raw second moment is

$$\mathbb{E}[X_i X_j] = \mu_i \mu_j + \mathcal{G}_{ij} + a_\varepsilon (\mu_i b_j + \mu_j b_i + \mathcal{G}_{ij} (c_i + c_j)) + a_\varepsilon^2 \Xi_{ij}. \quad (7.16)$$

Subtracting $m_\star m_\star^\top$ gives the exact connected two-point function. These identities follow from the Gaussian characteristic function and Stein's identity. Thus the numerical study retains an analytic multivariate ground truth while satisfying the exact source-domain assumptions.

Unless otherwise stated, $d = 8$. The random-number generator is NumPy's default generator with master seed 20260924. A raw dynamics matrix $A_{\text{raw}} \in \mathbb{R}^{d \times d}$ with independent $\mathcal{N}(0, 1)$ entries is generated once and

$$A = 0.72 A_{\text{raw}} / \rho(A_{\text{raw}}), \quad (7.17)$$

where $\rho(A_{\text{raw}})$ is the spectral radius. An independent standard-normal matrix is factorized by NumPy QR; if Q_U denotes the resulting orthogonal factor, then

$$U = Q_U \text{diag}(0.85, 0.85 + 0.30/(d-1), \dots, 1.15). \quad (7.18)$$

Next, $C = 0.78I + \Delta_C$ with $(\Delta_C)_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 0.06^2)$ generated once; $P_0 = 0.45^2 I$; $Q = 0.20^2(I + 0.20\mathbf{1}\mathbf{1}^\top/d)$; $R_y = 0.24^2(I + 0.15\mathbf{1}\mathbf{1}^\top/d)$; and $\beta = 0.55$. The ten trajectory realizations use deterministic seeds $20260924 + 100 + r$, $r = 0, \dots, 9$.

The amplitude has the weak-coupling expansion

$$a_\varepsilon = 1 - e^{-\varepsilon} = \varepsilon - \frac{1}{2}\varepsilon^2 + \frac{1}{6}\varepsilon^3 + O(\varepsilon^4), \quad (7.19)$$

so every coefficient in the complete order- p dressed expansion of m and G is available analytically from (7.15)–(7.16). The native perturbative sweep uses

$$\varepsilon \in \{0.001, 0.002, 0.004, 0.007, 0.01, 0.02, 0.04, 0.07, 0.10, 0.16, 0.25, 0.36\}, \quad (7.20)$$

whereas the baseline comparison uses

$$\varepsilon \in \{0.01, 0.02, 0.04, 0.07, 0.10, 0.16, 0.25, 0.36\}. \quad (7.21)$$

The exact self-energy localization sweep uses $\lambda \in \{0.25, 0.40, 0.60, 0.80\}$, with $\varepsilon = \lambda^2$. Figure 1 uses $\lambda = 0.60$, and Figures 3 and 5 use $\lambda = 0.80$. The log–log slopes in Figure 4 are fitted only to the first five native values, $0.001 \leq \varepsilon \leq 0.01$. The smallest fitted order-two mean and covariance errors are respectively 1.60×10^{-10} and 6.10×10^{-10} ; values below 10^{-13} would have been excluded from a fit.

For the baseline experiment ($T = 10$), mean-field Gaussian VB uses 20-point one-dimensional Gauss–Hermite quadrature, is initialized at the Laplace mean with marginal standard deviations from the Laplace covariance, and is optimized in $(m, \log s)$ by L-BFGS-B with maximum 80 iterations, function tolerance 10^{-10} and gradient tolerance 10^{-7} . The EP curve is explicitly a *12-sweep damped Gaussian EP calculation*, not a claimed EP fixed point: it uses 24-point one-dimensional Gauss–Hermite quadrature for unary tilted moments, a 10×10 product Gauss–Hermite rule for pair tilted moments, synchronous site proposals followed by a global damping factor 0.15, and exactly 12 sweeps, with no convergence claim or early-stopping tolerance. It is initialized at the $\varepsilon = 0$ Gaussian reference: the global natural parameters are P_u and $h_u = P_u \mu$, implemented by unary sites with $\tau_i = (P_u)_{ii}$ and $\eta_i = (h_u)_i$, pair sites carrying the nonzero off-diagonal entries of P_u , and zero additional nonlinear-site corrections. If a damped global update is not positive definite, the damping is repeatedly halved until it is admissible. The inverse state map required in each tilted integral is solved componentwise by Newton iteration. The Laplace mode is obtained by Newton iteration in the Gaussian latent coordinate with at most 60 iterations and stopping rule $\|\Delta u\| < 10^{-11}(1 + \|u\|)$. The oracle sampling curves use the seeds stated in [Sampling-based trajectory solutions](#) and independent exact draws from (7.12).

The timing experiment in Figure 6 is unchanged by the manufactured-coordinate replacement because it measures only the structured banded linear-algebra kernel. It uses SciPy’s symmetric banded solver on a Linux x86-64 virtual machine exposing five AMD EPYC 9V74 vCPUs, Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, JAX 0.9.0.1, Matplotlib 3.10.8 and OpenBLAS 0.3.30 (five BLAS threads). One warm-up solve is discarded before each recorded solve. Horizons are $T \in \{200, 400, 800, 1600, 3200, 6400\}$, with $d = 8$ and $R = 3$. These timings test only the scaling exponent of the structured solve, not implementation speed across methods.

7.1. Results supporting the native trajectory-field theorems

Figure 1 evaluates the exact bare Hessian $H_y[m_\star]$ and exact inverse connected two-point function $G_\star^{-1}$ for $T = 30$ and $\lambda = 0.60$. The plotted self-energy is then defined by the exact Dyson decomposition $\Sigma_\star = H_y[m_\star] - G_\star^{-1}$. The bare Hessian vanishes identically beyond lag one, while $G_\star^{-1}$ has nonzero off-band blocks; at lag two the normalized off-band block norm is approximately 4.32×10^{-6} . The equality between the off-band $G_\star^{-1}$ and $-\Sigma_\star$ curves is therefore not an independent numerical verification—it

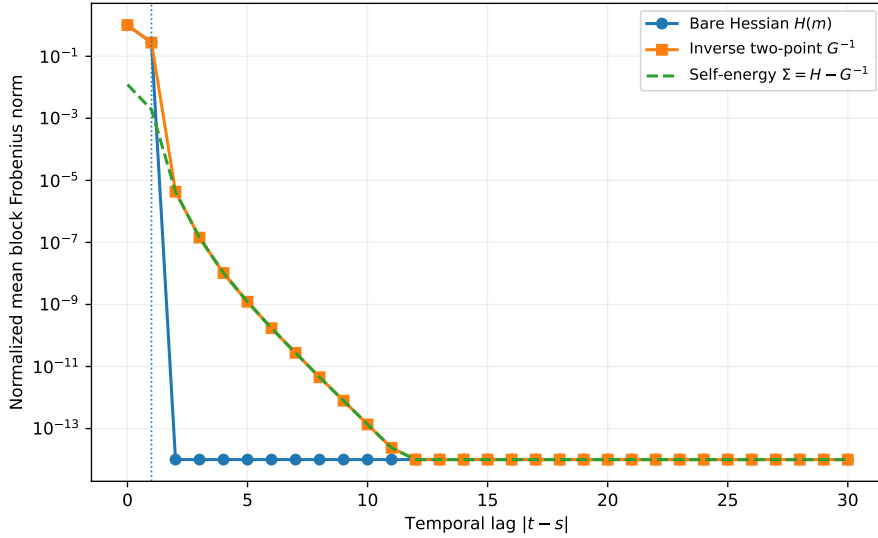


FIG. 1. Generated temporal nonlocality of the inverse connected two-point function. The bare Hessian is block tridiagonal and vanishes for $|t-s| > 1$, whereas G_*^{-1} develops an off-band tail. The self-energy is plotted from its defining Dyson decomposition $\Sigma_* = H_y[m_*] - G_*^{-1}$; coincidence of the two off-band curves is therefore an identity, not an independent numerical test. Curves are means over ten runs and bands show ± 1 empirical standard deviation.

is the defining decomposition in Theorem 5.1. The nontrivial numerical content is the presence and magnitude of the off-band inverse-two-point blocks themselves.

Figure 2 repeats the *exact* physical self-energy measurement for $\lambda \in \{0.25, 0.40, 0.60, 0.80\}$. To prevent subtraction/inversion roundoff in $\Sigma_* = H_y[m_*] - G_*^{-1}$ from determining the semilog slopes, only lags $2 \leq |t-s| \leq 10$ whose mean normalized self-energy norm is at least 10^{-11} are admitted to the fit. This retains lags 2–6 for $\lambda = 0.25, 0.40$ and lags 2–7 for $\lambda = 0.60, 0.80$, giving slopes -2.501 , -2.501 , -2.346 and -2.346 , respectively. Stronger nonlinearity raises the magnitude of the self-energy tail while preserving a clean exponential regime. Theorem 5.3 is a result for a fixed finite retained skeleton family represented by $\Sigma^{(L)}$, so this figure is not a direct numerical test of that theorem; rather, it displays the corresponding localization phenomenon for the exact manufactured self-energy.

Figure 3 applies the same range projection to the *exact* self-energy at $\lambda = 0.80$ and recomputes the covariance. The mean relative covariance error drops from 3.10×10^{-5} at $R = 1$ to 1.01×10^{-13} at $R = 10$. The semilog fit uses only $R = 1, \dots, 7$, for which the mean error is at least 10^{-11} ; the smaller $R = 8, 9, 10$ values are displayed but excluded from the fit as numerical-floor points. The resulting fitted slope is -2.298 , corresponding to a factor of approximately 0.100 for each additional retained lag. Corollary 5.2 is stated for a finite- L self-energy whose exponential decay follows from Theorem 5.3; Figure 3 instead demonstrates the identical resolvent/range mechanism for the exact self-energy, whose exponential tail is measured in Figure 2.

The perturbative-order result is tested directly in Figure 4. For $p = 0, 1, 2$ we retain every coefficient of the complete weak-coupling expansion through ε^p and compare with the exact transformed Gaussian moments. Proposition A.3 identifies these direct moment coefficients with the complete perturbative 2PI stationary branch coefficients. The fitted slopes use only $\varepsilon \in \{0.001, 0.002, 0.004, 0.007, 0.01\}$ and the numerical-floor rule stated above. The fitted mean-error slopes are 0.994, 1.995 and 2.995; the

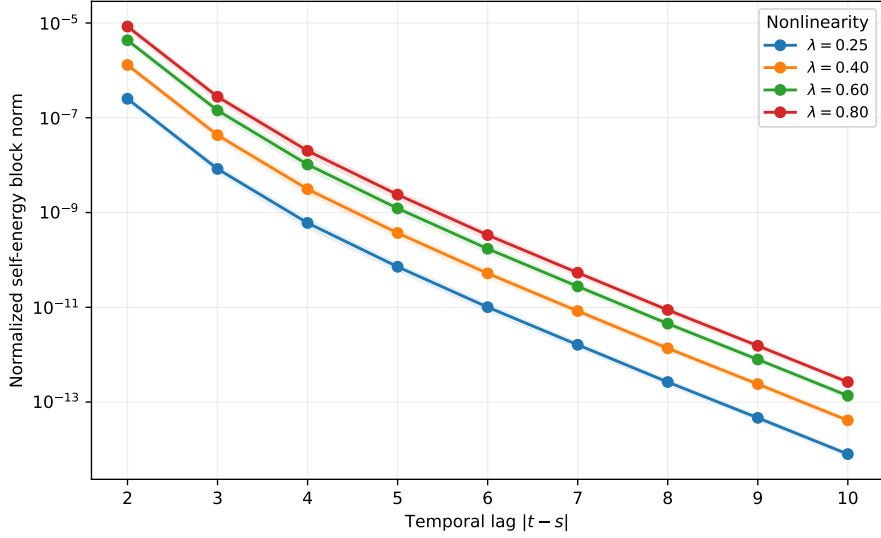


FIG. 2. Exponential localization of the *exact* generated self-energy for four nonlinearities. The finite-retained-family localization theorem concerns $\Sigma^{(L)}$; this exact-family plot demonstrates the corresponding decay mechanism rather than directly verifying that theorem. Each curve is the mean of ten exact runs; shading denotes ± 1 standard deviation.

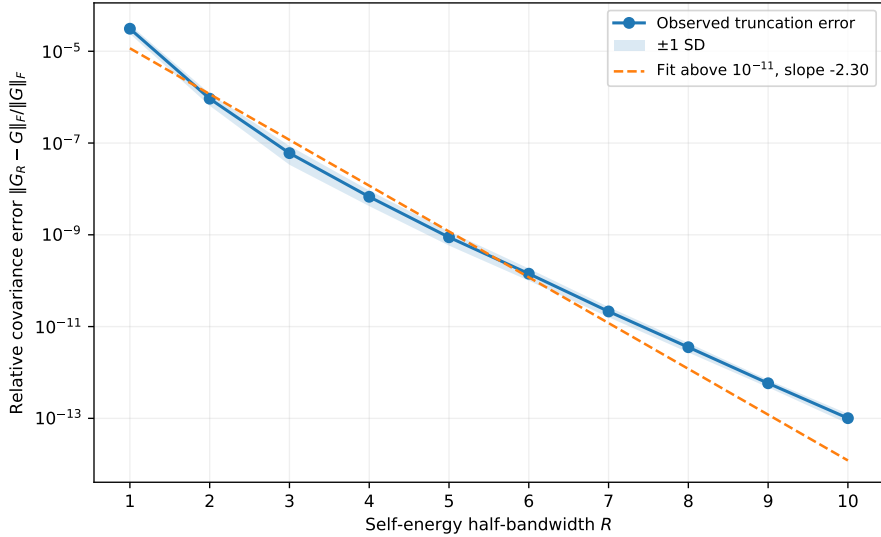


FIG. 3. Covariance error from range truncation of the exact self-energy. The exponential decay supports the same resolvent mechanism used in Corollary 5.2; the plotted self-energy itself does not arise from the finite retained skeleton family in Theorem 5.3.

covariance slopes are 0.993, 2.033 and 2.990. Thus the experiment reproduces the $O(\varepsilon^{p+1})$ conclusion of Theorem 5.2 without fitting the orders after the fact. At the strongest plotted coupling $\varepsilon = 0.36$,

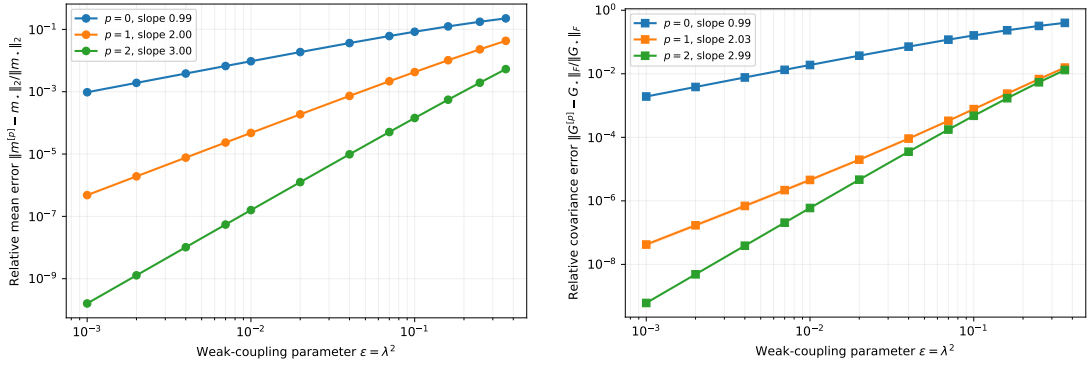


FIG. 4. Perturbative consistency of the dressed trajectory expansion. Left: trajectory-mean error. Right: connected two-point error. The fitted slopes are essentially $p + 1$ for $p = 0, 1, 2$.

the order-two relative mean and covariance errors are approximately 5.32×10^{-3} and 1.31×10^{-2} , respectively, on the $T = 30$ native-theorem experiment.

For the exact-family diagnostics define the projected exact physical self-energy

$$(\Sigma_\star^{(R)})_{ts} := \begin{cases} (\Sigma_\star)_{ts}, & |t - s| \leq R, \\ 0, & |t - s| > R. \end{cases} \quad (7.22)$$

Figure 5 makes a direct *pointwise* check of the sufficient inequality in Theorem 5.5 at the exact/reference point. For each run and range R , it reports $h = \lambda_{\min}(H_y[m_\star])$, $\|\Sigma_\star^{(R)}\|_2$, and the certified margin $h - \|\Sigma_\star^{(R)}\|_2$. The margin remains strictly positive for every run and every $R \in \{0, \dots, 10\}$; over the ten runs the smallest observed margin is greater than 4.60. This is not a uniform check over an iteration domain, nor is $\Sigma_\star^{(R)}$ identified with a finite- L self-energy; it verifies the theorem's pointwise spectral mechanism for the exact projected self-energy.

Finally, Figure 6 isolates the structured banded factorization/solve step in Theorem 5.6. With $d = 8$ and fixed range $R = 3$, the horizon is varied from $T = 200$ to $T = 6400$. Ten independent timing runs are used at each horizon. The fitted log-log slope is 1.06, close to the $O(T)$ prediction; the mean solve time increases from 0.425 ms to 17.6 ms over a factor of 32 in horizon.

7.2. Comparison with four baseline solution families

The second subsection uses the same nonlinear transformed state-space family but shortens the horizon to $T = 10$ so that the 12-sweep damped Gaussian EP calculations can be repeated over all nonlinearity levels and all ten realizations. Every method receives the same $d = 8$ conditioned target. The proposed curve is the complete order-two dressed weak-coupling series, denoted the *order-2 dressed series* below. The four baseline families are derived from the common posterior representation in the post-reference appendix, Appendix.

The variational baseline is fully factorized mean-field Gaussian variational Bayes, not general variational Bayes. Its expectations are evaluated with the quadrature and L-BFGS protocol specified above; all 80 optimization problems in the sweep converge. The EP baseline is the explicitly finite-iteration 12-sweep damped Gaussian EP calculation described above. It uses a full global Gaussian

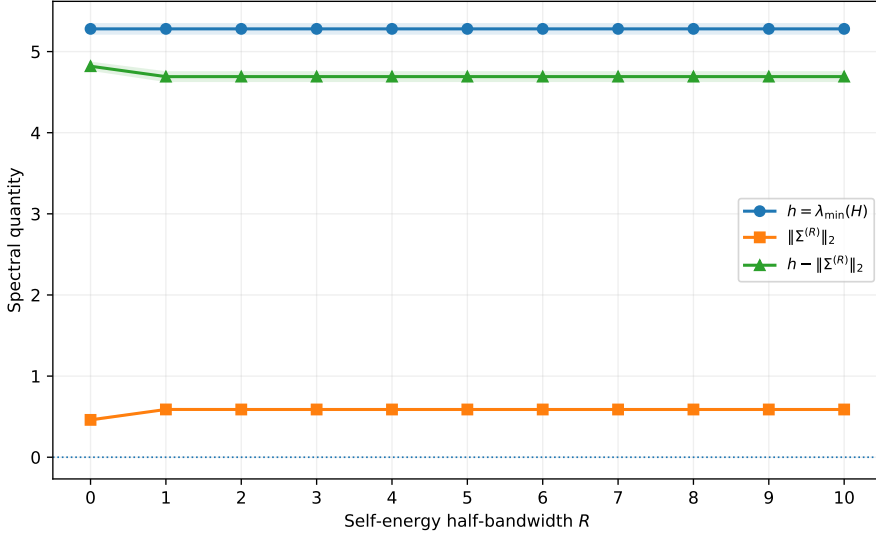


FIG. 5. Pointwise check of the sufficient spectral-admissibility condition at the exact/reference point. The curves report $h = \lambda_{\min}(H_y[m_*])$, $\|\Sigma_*^{(R)}\|_2$, and the certified margin $h - \|\Sigma_*^{(R)}\|_2$ over ten runs; bands show ± 1 standard deviation.

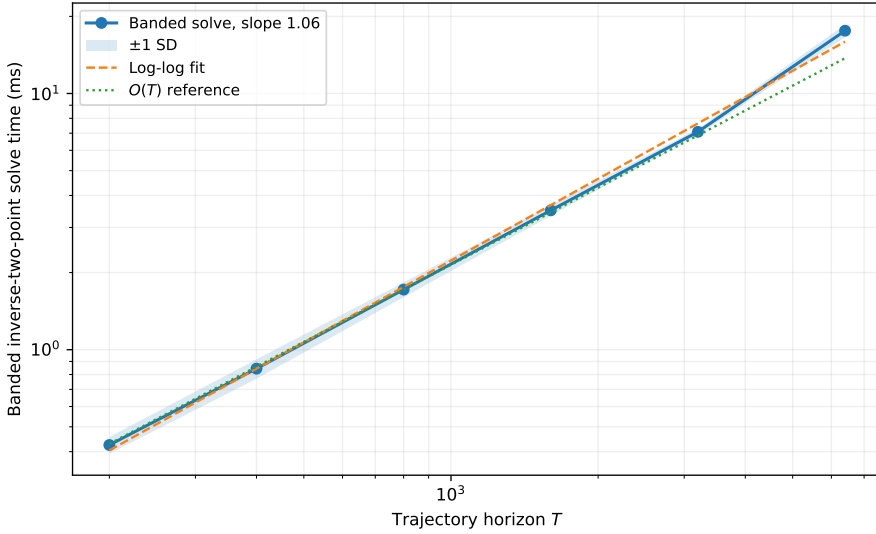


FIG. 6. Empirical complexity of a fixed-range block-banded dressed inverse-two-point solve. The measured log-log slope is 1.06, compared with the $O(T)$ reference.

approximation with scalar unary and pair Gaussian sites; it is not identified with the fixed point analysed later in [Expectation propagation](#). The Laplace baseline uses the trajectory mode and inverse action Hessian with the stated Newton stopping rule. The sampling baseline is a deliberately favourable oracle benchmark: it receives independent exact target draws and therefore eliminates mixing, proposal,

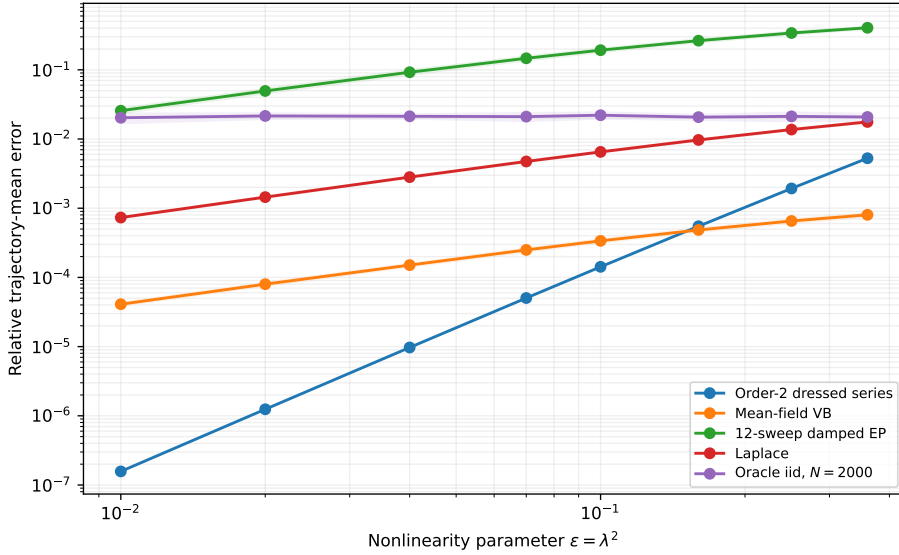


FIG. 7. Relative trajectory-mean error versus nonlinearity for the order-2 dressed series and four baseline solution families. Curves are means over ten runs; bands show ± 1 standard deviation.

resampling and weight-degeneracy effects. Figures 7–8 use $N = 2000$ such oracle draws, isolating the finite- N Monte Carlo limitation.

Figure 7 shows trajectory-mean error versus ε . At $\varepsilon = 0.36$, the order-2 dressed series gives 5.28×10^{-3} mean relative error, compared with 8.03×10^{-4} for mean-field VB, 4.05×10^{-1} for the 12-sweep EP calculation, 1.76×10^{-2} for Laplace and 2.08×10^{-2} for the oracle $N = 2000$ sampler. The purpose of the plot is not the single endpoint ordering: the order-2 dressed-series curve follows the third-order weak-coupling scaling predicted by Theorem 5.2. The remaining curves arise from different approximation structures; the post-reference appendix gives separately qualified results for the mean-field covariance restriction, Gaussian-EP fixed points, Laplace weak-coupling error and finite- N sampling. The explicitly finite 12-sweep EP calculation plotted here is not identified with the EP fixed point analysed in Expectation propagation.

The two-point comparison in Figure 8 is more discriminating. At $\varepsilon = 0.36$, the order-2 dressed series has relative covariance error 1.29×10^{-2} , compared with 4.61×10^{-1} for mean-field VB, 5.21×10^{-1} for the 12-sweep EP calculation, 2.14×10^{-2} for Laplace and 1.87×10^{-1} for 2000 independent exact samples. The large VB error is expected from its diagonal covariance restriction; [Mean-field Gaussian variational Bayes and its full-Gaussian parent](#) proves the resulting nonzero covariance-error floor whenever the true conditioned trajectory has off-diagonal correlation.

Figure 9 resolves the covariance by temporal lag at $\varepsilon = 0.36$. The order-two dressed approximation is visually indistinguishable from the exact reference over the displayed range. Mean-field VB has zero off-diagonal covariance by construction. The 12-sweep damped Gaussian EP calculation and Laplace recover the qualitative decay but exhibit systematic deviations. For the sampling covariance plots we use

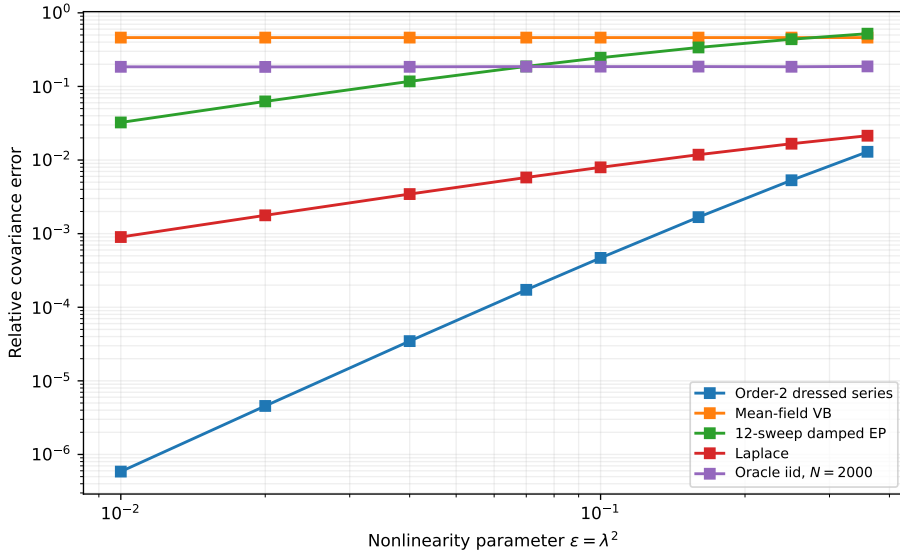


FIG. 8. Relative connected two-point error versus nonlinearity for the same five methods and the same ten conditioned trajectories.

the oracle-centred sample-average estimator

$$\hat{G}_N = \frac{1}{N} \sum_{i=1}^N (X^{(i)} - m_\star)(X^{(i)} - m_\star)^\top, \quad (7.23)$$

so each covariance entry is exactly an ordinary sample average to which Theorem A.3 applies with $f(X) = \text{vec}[(X - m_\star)(X - m_\star)^\top]$. Its Frobenius noise floor is nevertheless conspicuous in the small long-lag blocks: at lag five its normalized block norm is approximately 6.14×10^{-2} , compared with the exact 4.97×10^{-3} .

The sampling comparison is completed by varying its budget. Figure 10 uses between 50 and 20000 exact independent trajectory draws and compares the resulting trajectory-mean error with the fixed deterministic approximations. The oracle sampling curve has fitted slope -0.503 , essentially the canonical $N^{-1/2}$ rate. Even at $N = 20000$, its mean relative error is 6.50×10^{-3} on this 88-dimensional path, while the order-two dressed error is 5.28×10^{-3} .

Figure 11 repeats the sample-count experiment for the full covariance using the oracle-centred estimator (7.23). The oracle sampling slope is -0.502 . At $N = 20000$ its relative covariance error remains 5.89×10^{-2} , compared with 1.29×10^{-2} for the order-2 dressed series. [Sampling-based trajectory solutions](#) shows that this is not an artefact of the selected sampler: for the ordinary sample mean, even independent exact draws have mean-square error proportional to N^{-1} , leading to an explicit finite- N sample-count regime in which a controlled deterministic approximation has lower error. This is a statistical sample-budget statement, not a wall-clock or computational-budget comparison.

8. Discussion

The field-theoretic formulation separates three issues that are often merged in nonlinear trajectory inference. The first is exact structure. Equations (3.1)–(3.6) and the exact 2PI stationarity equations

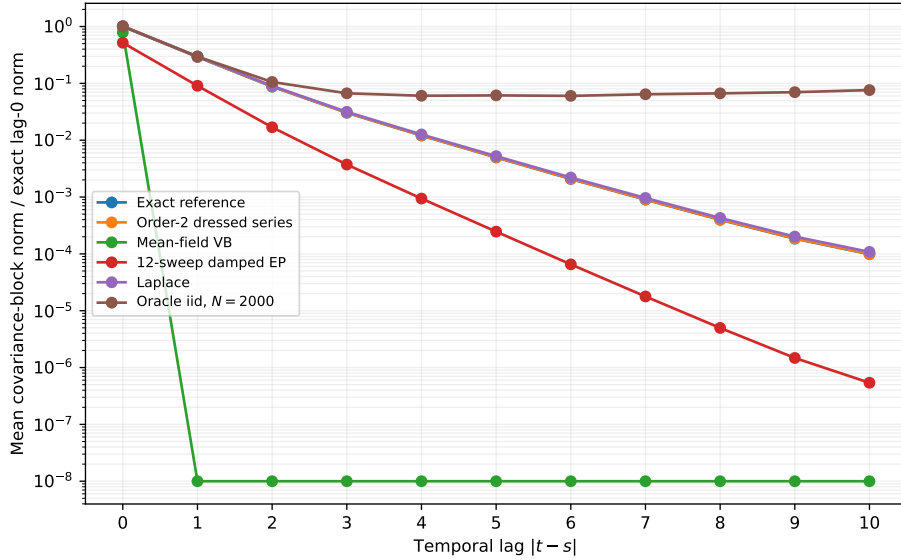


FIG. 9. Temporal covariance profile at $\varepsilon = 0.36$. The mean-field VB line is clipped to the plotting floor for nonzero lags because its off-diagonal covariance is exactly zero.

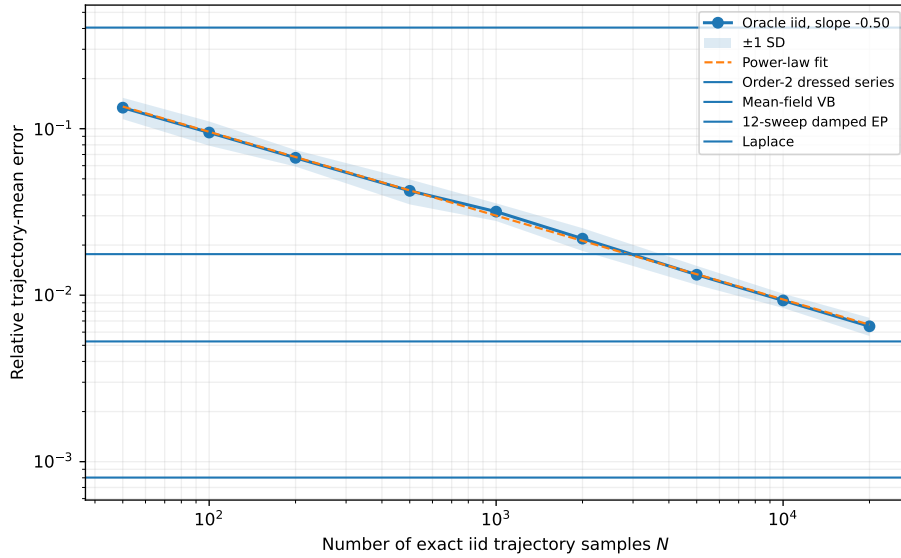


FIG. 10. Finite-sample-count trajectory-mean comparison at $\varepsilon = 0.36$. The sampling method is an oracle receiving independent exact target draws; its fitted slope is -0.503 .

contain no approximation. The second is truncation structure. Theorems 5.2–5.5 identify explicit conditions under which perturbative order, the retained finite skeleton family, or temporal range is

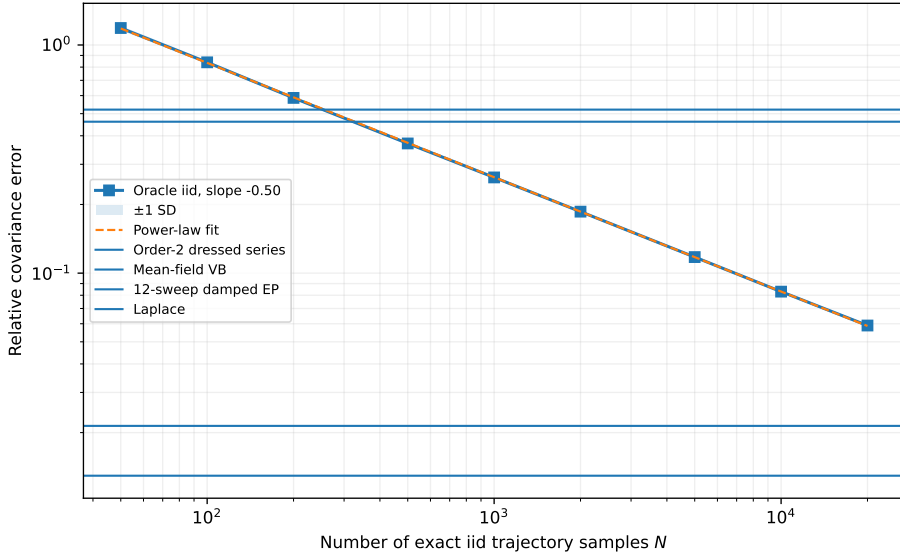


FIG. 11. Finite-sample-count connected two-point comparison at $\varepsilon = 0.36$. The oracle independent-sampling covariance error has fitted slope -0.502 .

meaningful. The third is numerical solution. Damping, Newton steps and stopping residuals concern how a chosen truncated stationary point is reached; they do not certify the truncation itself.

The generated-nonlocality result is the main structural distinction. A local nonlinear stochastic model has a block-tridiagonal bare Hessian, but the exact inverse connected two-point function can acquire off-band interaction through Σ . This gives the self-energy a concrete, coordinate-dependent dynamical interpretation: it measures fluctuation-generated coupling in the two-point response of the chosen state field. It does not alter the nearest-neighbour Markov factorization of the non-Gaussian path density. For a genuinely quadratic action, the exact interaction functional satisfies $\Phi = 0$ and hence $\Sigma = 0$ identically, so $G^{-1} = H$ is block banded. Away from that limit, forcing the inverse connected two-point function to remain local is an approximation to two-point structure, not a consequence of the Markov factorization.

The experiments intentionally use an exactly solvable nonlinear coordinate family rather than a collection of real-data benchmarks. That choice has two advantages for the present paper. First, it makes the theorem-predicted powers and exponential tails directly testable against exact moments. Second, every baseline is evaluated against the same target without tuning a separate data-generating process to favour a particular method. The limitation is equally clear: the numerical section is evidence for the theoretical mechanisms, not a claim that the manufactured family spans all practical nonlinear or multimodal state-space problems.

The order-two numerical curve should also be interpreted precisely. It is the second-order weak-coupling Taylor polynomial obtained from (7.19) by evaluating all required Gaussian contractions. Proposition A.3 shows that its coefficients coincide through $O(\varepsilon^2)$ with those of the stationary branch generated by the complete order-two perturbative 2PI functional. It is therefore a direct realization of the perturbative effective-action logic at the level of Taylor coefficients, but it is not the finite- ε self-consistent solution of the truncated functional and is not presented as a benchmark of a fully general-purpose

self-consistent 2PI software implementation. For an arbitrary action, the retained skeletons must be assembled from the local derivative tensors and the coupled equations (6.2)–(6.3) solved numerically.

Several extensions follow naturally. Higher n PI effective actions would promote higher connected functions to independent variables when the two-point closure is insufficient. Continuous-time limits can be developed after the finite-grid identities are established and the limiting functional measure is specified. Symmetries of the action can be propagated to Ward-type constraints on the truncated effective action. Finally, the self-energy range R can be selected adaptively from a posteriori tail norms rather than fixed in advance, provided that the range projection is kept distinct from the skeleton-order approximation.

9. Conclusion

A conditioned nonlinear stochastic trajectory admits an exact field theory on its finite time grid. The bare action is local, the Schwinger–Dyson hierarchy is exact, and the 2PI effective action yields a dressed trajectory mean and connected two-point function through coupled stationarity equations. The resulting Dyson decomposition reveals that off-band structure of the inverse connected two-point function is fluctuation generated in the chosen state coordinates: beyond the bare nearest-neighbour Hessian band, G^{-1} is exactly the negative self-energy. This statement is distinct from, and does not modify, the nearest-neighbour Markov factorization of the non-Gaussian path density.

This structure leads to a controlled approximation programme rather than a single fixed smoother. The retained skeleton family controls which interaction processes are represented; self-energy range controls how much generated temporal interaction is retained. Under explicit assumptions, complete perturbative order- p closures achieve $O(\varepsilon^{p+1})$ local-in- ε , branchwise error near a nondegenerate reference point, and self-energies from fixed finite retained skeleton families inherit exponential temporal localization. At fixed closure inputs, range projection gives exponentially small spectral-norm covariance-update error; under the additional Jacobian-stability hypothesis of Theorem 5.4, the re-solved range-truncated branch is exponentially close. Positive dressed inverse-two-point operators remain admissible, and fixed-range banded factorization and solves scale linearly with trajectory length. The numerical results reproduce the perturbative orders, exhibit exact-family self-energy localization and range decay, verify the pointwise admissibility inequality on the manufactured family, and confirm linear-in-horizon banded-solve scaling. Figures 2–3 concern the exact physical self-energy and are not presented as direct tests of the finite- L localization theorem or the re-solved branch-stability theorem. The post-reference appendix (Appendix) shows how the same target is treated by four standard Bayesian solution families and identifies precisely qualified regimes in which their structural, perturbative or finite- N errors separate from the dressed trajectory approximation.

Data availability

All numerical data in this paper are synthetic. The data-generating equations, all fixed parameter values and the random seed are specified in Section 7; the exact reference moments are given in closed form in (7.15)–(7.16) and the post-reference subsection [Exact moments and perturbative coefficients for the manufactured family](#).

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Conflict of interest

The author declares no conflict of interest.

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Appendix

Bayesian recasting of the common trajectory problem

The main text requires only the conditioned path weight (2.1). To compare with standard approximate-inference methods, normalize that weight and use Bayes' rule. The posterior trajectory density is

$$\pi_y(x_{0:T}) := p(x_{0:T} | y_{0:T}) = \frac{1}{Z_y[0]} e^{-S_y[x]}. \quad (\text{A.1})$$

Thus the methods developed in [Mean-field Gaussian variational Bayes and its full-Gaussian parent](#), [Expectation propagation](#), [Laplace trajectory smoothing](#), and [Sampling-based trajectory solutions](#) target exactly the same object as the trajectory field theory. The distinction is not the target distribution but the representation used to approximate its expectations and connected responses.

For later use, let

$$z_\star := (m_\star, G_\star) \quad (\text{A.2})$$

collect the exact posterior mean and covariance. A weak-coupling family is written

$$S_\varepsilon(x) = S_0(x) + \varepsilon V_1(x) + O(\varepsilon^2), \quad S_0(x) = \frac{1}{2}(x - m_0)^\top G_0^{-1}(x - m_0) + \text{const}, \quad (\text{A.3})$$

whenever a quadratic reference is used for a first-order asymptotic comparison, with the remainder controlled in the differentiability class required below.

Mean-field Gaussian variational Bayes and its full-Gaussian parent

Variational inference approximates (A.1) by a member of a tractable family and minimizes a Kullback–Leibler divergence; see, for example, [16]. For a Gaussian trial law

$$q(x) = \mathcal{N}(x; m, G_q), \quad (\text{A.4})$$

the variational free energy, up to a constant independent of q , is

$$\mathcal{F}_{\text{VB}}(m, G_q) = \mathbb{E}_q[S_y(X)] - \frac{1}{2} \log \det G_q. \quad (\text{A.5})$$

When G_q is unconstrained, Gaussian differentiation identities give the stationary conditions

$$\mathbb{E}_q[\nabla S_y(X)] = 0, \quad (\text{A.6})$$

$$G_q^{-1} = \mathbb{E}_q[\nabla^2 S_y(X)]. \quad (\text{A.7})$$

These equations may coincide with a low-order Gaussian or Hartree closure in special models, but the logical construction is different: VB restricts the admissible probability law to the chosen Gaussian family, whereas the exact 2PI construction follows from a double Legendre transform.

The numerical baseline in Section 7.2 is the stricter mean-field Gaussian family

$$q_{\text{MF}}(x) = \prod_{a=1}^n \mathcal{N}(x_a; m_a, s_a^2), \quad G_q = \text{diag}(s_1^2, \dots, s_n^2). \quad (\text{A.8})$$

For the transformed experimental action, the expectations in (A.5) are evaluated coordinatewise by Gauss–Hermite quadrature and the variables $(m, \log s)$ are optimized directly.

Small- ε covariance separation for mean-field Gaussian VB

Let Diag denote the linear subspace of diagonal $n \times n$ matrices, and let $\text{Off}(G) := G - \text{diag}(G)$. The next result identifies a representation error that no optimization of (A.8) can remove.

Theorem A.1 (Mean-field covariance-error floor) *For any exact covariance $G_\star$ and any mean-field Gaussian covariance $D_{\text{MF}} \in \text{Diag}$,*

$$\|D_{\text{MF}} - G_\star\|_F \geq \|\text{Off}(G_\star)\|_F. \quad (\text{A.9})$$

Equality is attained by $D_{\text{MF}} = \text{diag}(G_\star)$. Consequently, suppose $G_\star(\varepsilon)$ is continuous at $\varepsilon = 0$ and

$$\delta_0 := \|\text{Off}(G_\star(0))\|_F > 0. \quad (\text{A.10})$$

If a complete order- p dressed approximation satisfies

$$\left\| G^{[p]}(\varepsilon) - G_\star(\varepsilon) \right\|_F \leq C|\varepsilon|^{p+1}, \quad (\text{A.11})$$

then there exists $\varepsilon_0 > 0$ such that for $0 < |\varepsilon| < \varepsilon_0$,

$$\left\| G^{[p]}(\varepsilon) - G_\star(\varepsilon) \right\|_F < \|G_{\text{MF}}(\varepsilon) - G_\star(\varepsilon)\|_F \quad (\text{A.12})$$

for every mean-field Gaussian approximation $G_{\text{MF}}(\varepsilon)$.

Proof The diagonal and off-diagonal subspaces are orthogonal in the Frobenius inner product. Therefore

$$\|D_{\text{MF}} - G_\star\|_F^2 = \|D_{\text{MF}} - \text{diag}(G_\star)\|_F^2 + \|\text{Off}(G_\star)\|_F^2, \quad (\text{A.13})$$

which proves (A.9). By continuity, $\|\text{Off}(G_\star(\varepsilon))\|_F \geq \delta_0/2$ for sufficiently small $|\varepsilon|$. Choose ε_0 further so that $C|\varepsilon|^{p+1} < \delta_0/2$. Then (A.11) is strictly smaller than the lower bound on every mean-field covariance error. $\square$

For a coupled state trajectory, condition (A.10) already holds in the Gaussian reference problem whenever the chain has nonzero temporal or cross-component covariance. This explains why the covariance error of the numerical mean-field baseline in Figure 8 does not vanish as $\varepsilon \rightarrow 0$.

Expectation propagation

Expectation propagation (EP) replaces difficult factors by tractable sites and repeatedly matches moments of locally tilted distributions [17]. Write the posterior factorization abstractly as

$$\pi_y(x) \propto \prod_{a \in \mathcal{A}} f_a(x_a), \quad (\text{A.14})$$

where x_a denotes the subset of trajectory variables touched by factor a . Gaussian EP introduces sites

$$\tilde{f}_a(x_a) \propto \exp\left(-\frac{1}{2}x_a^\top \Lambda_a x_a + \eta_a^\top x_a\right) \quad (\text{A.15})$$

so that

$$q_{\text{EP}}(x) \propto \prod_{a \in \mathcal{A}} \tilde{f}_a(x_a) \quad (\text{A.16})$$

is a global Gaussian.

For one site, define the cavity law

$$q^{\setminus a}(x_a) \propto \frac{q_{\text{EP}}(x_a)}{\tilde{f}_a(x_a)} \quad (\text{A.17})$$

and the tilted law

$$\hat{p}_a(x_a) \propto q^{\setminus a}(x_a) f_a(x_a). \quad (\text{A.18})$$

The site is updated so that the corresponding marginal of the new global Gaussian has the same first two moments as (A.18). Iterating these cavity–tilted–projection steps produces an EP fixed point. Related expectation-consistent free-energy formulations are described in [18].

For the experimental family, the transformed Gaussian action decomposes into scalar Jacobian/unary factors and scalar pair factors associated with nonzero entries of the block-banded base precision. The implemented numerical baseline retains a full global Gaussian, evaluates unary tilted moments by one-dimensional Gauss–Hermite quadrature and pair tilted moments by product Gauss–Hermite quadrature, and applies exactly 12 damped synchronous sweeps. We therefore call it the *12-sweep damped Gaussian EP calculation*; it is not asserted to have reached the EP fixed point defined above.

Gaussian cumulant obstruction and a conditional second-order separation criterion

A Gaussian EP fixed point can encode a dense covariance, but its connected cumulants of order three and above vanish identically. Let $\kappa_\star^{(r)}$ denote the exact connected r th-order trajectory cumulant and let $\kappa_{\text{EP}}^{(r)}$ denote the cumulant of the global Gaussian EP approximation.

Proposition A.1 (Gaussian EP cumulant obstruction) *For every $r \geq 3$,*

$$\kappa_{\text{EP}}^{(r)} = 0. \quad (\text{A.19})$$

Therefore, if $\kappa_\star^{(r)} \neq 0$ for any $r \geq 3$,

$$\left\| \kappa_{\text{EP}}^{(r)} - \kappa_\star^{(r)} \right\| = \left\| \kappa_\star^{(r)} \right\|. \quad (\text{A.20})$$

Proof The cumulant generating function of a Gaussian law is a polynomial of degree two in the source. All derivatives of order three and above therefore vanish. $\square$

The obstruction does not imply that EP must have an error in every first or second moment. A sharper comparison for those quantities requires identifying the first omitted cumulant correction that feeds back into them. Perturbative analyses of Gaussian latent-variable EP show explicitly how corrections are built from the higher cumulants neglected by local moment matching, and in the corresponding expansion EP is correct at the leading order while higher-order corrections remain [19].

Proposition A.2 (Conditional second-order separation criterion for Gaussian EP) *Let z denote any fixed finite collection of source derivatives, for example (m, G) . Suppose an analytic weak-coupling*

family has expansions

$$z_*(\varepsilon) = z_0 + z_1\varepsilon + z_2\varepsilon^2 + O(\varepsilon^3), \quad (\text{A.21})$$

$$z_{\text{EP}}(\varepsilon) = z_0 + z_1\varepsilon + \tilde{z}_2\varepsilon^2 + O(\varepsilon^3), \quad (\text{A.22})$$

with $z_2 \neq \tilde{z}_2$, while a complete order-two dressed closure satisfies

$$z_{\text{dress},2}(\varepsilon) = z_0 + z_1\varepsilon + z_2\varepsilon^2 + O(\varepsilon^3). \quad (\text{A.23})$$

Then

$$\frac{\|z_{\text{dress},2}(\varepsilon) - z_*(\varepsilon)\|}{\|z_{\text{EP}}(\varepsilon) - z_*(\varepsilon)\|} \longrightarrow 0 \quad \text{as } \varepsilon \rightarrow 0. \quad (\text{A.24})$$

Proof Equations (A.21)–(A.23) give

$$z_{\text{dress},2} - z_* = O(\varepsilon^3), \quad (\text{A.25})$$

$$z_{\text{EP}} - z_* = (\tilde{z}_2 - z_2)\varepsilon^2 + O(\varepsilon^3). \quad (\text{A.26})$$

Because the leading EP coefficient is nonzero, the denominator is $\Theta(\varepsilon^2)$, so the ratio is $O(\varepsilon)$. $\square$

Proposition A.2 is deliberately conditional on a nonzero first omitted correction and is therefore a general algebraic criterion, not an unconditional dominance theorem for EP. In particular, this subsection does not derive $\tilde{z}_2 \neq z_2$ for the factorization used in Section 7.2; EP can also be exact in special Gaussian or symmetry-protected cases. Figures 7–9 are therefore independent finite- ε comparisons of the explicitly labelled 12-sweep damped EP calculation and are not evidence for, or a verification of, Proposition A.2.

Laplace trajectory smoothing

A trajectory Laplace approximation is defined locally about a nondegenerate mode branch. For a weak-coupling family, let $\hat{x}_L(\varepsilon)$ satisfy

$$\nabla S_\varepsilon(\hat{x}_L(\varepsilon)) = 0, \quad \hat{x}_L(0) = m_0, \quad \nabla^2 S_\varepsilon(\hat{x}_L(\varepsilon)) \succ 0 \quad (\text{A.27})$$

on the local branch under consideration. The corresponding Gaussian approximation is

$$q_{L,\varepsilon}(x) = \mathcal{N}(\hat{x}_L(\varepsilon), [\nabla^2 S_\varepsilon(\hat{x}_L(\varepsilon))]^{-1}). \quad (\text{A.28})$$

We write $m_L(\varepsilon) := \hat{x}_L(\varepsilon)$ and $G_L(\varepsilon) := [\nabla^2 S_\varepsilon(\hat{x}_L(\varepsilon))]^{-1}$ for the corresponding Laplace mean and covariance.

This is the trajectory analogue of the classical Laplace method for posterior moments [20]; Gaussian filtering and smoothing treatments are surveyed in [21].

For the experimental family, the mode is solved in the Gaussian latent coordinate u , including the transformation Jacobian. The reported state covariance is then the inverse Hessian with respect to the physical x coordinates, so the baseline is not given an artificially linearized observation model.

Observable-wise weak-coupling separation from Laplace

For the reference family (A.3), let $X \sim \mathcal{N}(m_0, G_0)$ under $\varepsilon = 0$ and assume V_1 and the $O(\varepsilon^2)$ remainder are sufficiently smooth and integrable for the following first-order expansions.

Theorem A.2 (First-order Laplace mismatch) *The exact mean and covariance satisfy*

$$m_\star(\varepsilon) = m_0 - \varepsilon G_0 \mathbb{E}_0[\nabla V_1(X)] + O(\varepsilon^2), \quad (\text{A.29})$$

$$G_\star(\varepsilon) = G_0 - \varepsilon G_0 \mathbb{E}_0[\nabla^2 V_1(X)] G_0 + O(\varepsilon^2). \quad (\text{A.30})$$

The Laplace approximation satisfies

$$m_L(\varepsilon) = m_0 - \varepsilon G_0 \nabla V_1(m_0) + O(\varepsilon^2), \quad (\text{A.31})$$

$$G_L(\varepsilon) = G_0 - \varepsilon G_0 \nabla^2 V_1(m_0) G_0 + O(\varepsilon^2). \quad (\text{A.32})$$

Hence, if either

$$\mathbb{E}_0[\nabla V_1(X)] \neq \nabla V_1(m_0) \quad (\text{A.33})$$

or

$$\mathbb{E}_0[\nabla^2 V_1(X)] \neq \nabla^2 V_1(m_0), \quad (\text{A.34})$$

then the corresponding Laplace error is $\Theta(\varepsilon)$, whereas a complete first-order dressed closure has error $O(\varepsilon^2)$. In that observable,

$$\frac{\text{Err}_{\text{dress},1}(\varepsilon)}{\text{Err}_L(\varepsilon)} \rightarrow 0. \quad (\text{A.35})$$

Proof For any observable f ,

$$\left. \frac{d}{d\varepsilon} \mathbb{E}_\varepsilon[f(X)] \right|_{\varepsilon=0} = -\text{Cov}_0(f(X), V_1(X)). \quad (\text{A.36})$$

Taking $f(X) = X$ and applying Gaussian Stein's identity gives

$$\text{Cov}_0(X, V_1(X)) = G_0 \mathbb{E}_0[\nabla V_1(X)], \quad (\text{A.37})$$

which yields (A.29). Applying the second-order Gaussian integration-by-parts identity to the centered quadratic observable gives (A.30).

The local Laplace mode branch solves

$$G_0^{-1}(\hat{x}_L(\varepsilon) - m_0) + \varepsilon \nabla V_1(\hat{x}_L(\varepsilon)) + O(\varepsilon^2) = 0. \quad (\text{A.38})$$

Expansion around m_0 gives (A.31). The Laplace precision along this branch is

$$G_0^{-1} + \varepsilon \nabla^2 V_1(\hat{x}_L(\varepsilon)) + O(\varepsilon^2), \quad (\text{A.39})$$

and a first-order matrix inverse expansion gives (A.32). Under either nondegeneracy condition, the corresponding difference between exact and Laplace coefficients at order ε is nonzero. Theorem 5.2 gives $O(\varepsilon^2)$ error for a complete first-order dressed closure, proving (A.35). $\square$

The theorem identifies the mechanism: Laplace uses derivatives evaluated at one trajectory, while the exact first correction depends on fluctuation averages of those derivatives. Equality can occur for special polynomial structures, but it is not generic.

Sampling-based trajectory solutions

Sampling methods approximate expectations under (A.1) rather than replacing the target by a deterministic parametric family. Sequential Monte Carlo methods propagate weighted particles through the state dynamics [22]; particle MCMC embeds such particle systems in invariant Markov kernels on trajectories [23].

Sequential and particle smoothing constructions

For a proposal $q_t(x_t | x_{t-1}, y_t)$, a particle filter generates

$$x_t^{(i)} \sim q_t(\cdot | x_{t-1}^{(a_i)}, y_t) \quad (\text{A.40})$$

with incremental importance weight

$$\tilde{w}_t^{(i)} \propto w_{t-1}^{(a_i)} \frac{M_t(x_t^{(i)} | x_{t-1}^{(a_i)}) \Lambda_t(y_t | x_t^{(i)})}{q_t(x_t^{(i)} | x_{t-1}^{(a_i)}, y_t)}. \quad (\text{A.41})$$

Normalized weights are used for resampling or for a subsequent backward-smoothing construction. Path-space MCMC instead specifies a kernel

$$K(x, dx') \quad (\text{A.42})$$

that leaves π_y invariant, for example through Metropolis–Hastings, Hamiltonian updates or particle Gibbs. In the limit of sufficient effective samples, these approaches are not biased by a finite deterministic closure. Their limitation at a fixed budget is Monte Carlo variance, together with whatever dependence or weight degeneracy the practical scheme introduces.

The numerical sampling baseline in Section 7.2 removes the latter difficulties entirely. It draws independent exact trajectories from the known reference law (7.12) and transforms them through (7.4). It is therefore a favourable iid exact-sampling benchmark that removes mixing, proposal, resampling and weight-degeneracy effects. For the $N = 2000$ baseline sweeps in Figures 7–9, the NumPy generator seed is $20260924 + 900000 + 1000r + j$, where $r = 0, \dots, 9$ indexes the trajectory realization and $j = 0, \dots, 7$ indexes the ordered grid (7.21). The sample-count mean experiment in Figure 10 uses seed $20260924 + 700000 + r$ for each run, and the oracle-centred covariance experiment in Figure 11 uses $20260924 + 810000 + r$.

Finite- N MSE separation from independent exact sampling

Theorem A.3 (Finite- N deterministic MSE separation) *Let $f(X) \in \mathbb{R}^k$ be an observable with finite covariance V_f under the exact conditioned path law, and let*

$$\theta_f := \mathbb{E}[f(X)]. \quad (\text{A.43})$$

For N independent exact trajectory draws, the ordinary Monte Carlo estimator

$$\hat{\theta}_{f,N} = \frac{1}{N} \sum_{i=1}^N f(X^{(i)}) \quad (\text{A.44})$$

satisfies

$$\mathbb{E} \left\| \hat{\theta}_{f,N} - \theta_f \right\|_2^2 = \frac{\text{Tr}(V_f)}{N}. \quad (\text{A.45})$$

Suppose a deterministic order- p dressed approximation obeys

$$\left\| \theta_f^{[p]} - \theta_f \right\|_2 \leq C_f |\varepsilon|^{p+1}. \quad (\text{A.46})$$

Then whenever

$$N < \frac{\text{Tr}(V_f)}{C_f^2 |\varepsilon|^{2p+2}}, \quad (\text{A.47})$$

the deterministic squared error is strictly smaller than the mean squared error of the independent exact-sampling estimator. This is a comparison at fixed sample count N and is not a statement about wall-clock computational cost.

Proof Independence gives

$$\text{Cov}(\hat{\theta}_{f,N}) = \frac{V_f}{N}. \quad (\text{A.48})$$

The estimator is unbiased, so its squared Euclidean mean error equals the trace of this covariance, proving (A.45). Equation (A.46) bounds the deterministic squared error by $C_f^2 |\varepsilon|^{2p+2}$. Inequality (A.47) makes this smaller than (A.45). $\square$

For the trajectory mean, $f(X) = X$ and $V_f = G_*$, so the threshold is explicit in the total trajectory variance. For the oracle-centred covariance estimator (7.23), take $f(X) = \text{vec}[(X - m_*)(X - m_*)^\top]$; then the same theorem applies componentwise and in vectorized Frobenius norm because the exact mean is held fixed in the estimator. For a stationary MCMC estimator satisfying a central limit theorem, V_f in (A.45) is replaced asymptotically by the integrated autocovariance matrix, typically increasing the constant. Theorem A.3 therefore uses a favourable iid exact-sampling benchmark for the sampling side. It does not claim dominance as $N \rightarrow \infty$; independent exact sampling eventually converges to the exact observable.

Exact moments and perturbative coefficients for the manufactured family

Let $u \sim \mathcal{N}(\mu, \mathcal{G})$ denote the stacked conditional Gaussian field, let $P_u = \mathcal{G}^{-1}$, and write

$$x = u + a_\varepsilon \sin u, \quad a_\varepsilon = 1 - e^{-\varepsilon}, \quad (\text{A.49})$$

componentwise. Since $|a_\varepsilon| < 1$ for $\varepsilon \in (-\log 2, \infty)$, (A.49) is a diffeomorphism on an open two-sided neighbourhood of $\varepsilon = 0$; the numerical experiments use only $\varepsilon = \lambda^2 \geq 0$. Its inverse has the weak-coupling expansion

$$u_\varepsilon(x) = x - \varepsilon \sin x + \varepsilon^2 \sin x \left(\cos x + \frac{1}{2} \right) + O(\varepsilon^3). \quad (\text{A.50})$$

Up to an x -independent constant, the exact conditioned action is

$$S_\varepsilon(x) = \frac{1}{2} (u_\varepsilon(x) - \mu)^\top P_u (u_\varepsilon(x) - \mu) + \sum_i \log(1 + a_\varepsilon \cos u_{\varepsilon,i}(x_i)). \quad (\text{A.51})$$

Writing $r = x - \mu$, $s := \sin x$ and $c := \cos x$ componentwise gives

$$S_\varepsilon(x) = S_0(x) + \varepsilon V_1(x) + \varepsilon^2 V_2(x) + O(\varepsilon^3), \quad (\text{A.52})$$

$$V_1(x) = -s^\top P_u r + \sum_i c_i, \quad (\text{A.53})$$

$$V_2(x) = \left[s \odot \left(c + \frac{1}{2} \mathbf{1} \right) \right]^\top P_u r + \frac{1}{2} s^\top P_u s + \sum_i \left(s_i^2 - \frac{1}{2} c_i - \frac{1}{2} c_i^2 \right). \quad (\text{A.54})$$

Thus the first two weak-coupling interaction functions, and hence the complete order-one and order-two 2PI coefficients, are explicitly determined by the manufactured family rather than inferred from the output curves. To display those coefficients without expanding every coordinate contraction, let $\phi \sim \mathcal{N}(0, G)$ and remove from each interaction function the pieces already carried by $S[m]$ and $H[m]$:

$$\tilde{V}_r(m, \phi) := V_r(m + \phi) - V_r(m) - \nabla V_r(m)^\top \phi - \frac{1}{2} \phi^\top \nabla^2 V_r(m) \phi, \quad r = 1, 2. \quad (\text{A.55})$$

Because V_1 and V_2 are finite sums of polynomial factors multiplied by sine and cosine, their Taylor expansions are absolutely integrable under the Gaussian measure; Gaussian exponential integrability therefore justifies termwise Wick contraction and rearrangement of the resulting series. For a Wick pairing appearing in a Gaussian contraction at fixed (m, G) , form its contraction multigraph: interaction vertices are graph vertices and every paired field occurrence is an edge carrying the corresponding propagator entry. Define $\mathcal{P}_{2\text{PI}}$ linearly on the Wick expansion by retaining a pairing if and only if its contraction multigraph is connected and remains connected after deletion of any two propagator edges, and by annihilating all other pairings. Thus $\mathcal{P}_{2\text{PI}}$ is a projection on Wick pairings, not an additional analytic approximation; it is the finite-dimensional combinatorial realization of the standard 2PI skeleton selection in [3]. If $\text{WickCov}_G(A, B)$ denotes the Wick expansion of $\mathbb{E}_G[AB] - \mathbb{E}_G[A]\mathbb{E}_G[B]$, define

$$\text{Cov}_G^{2\text{PI}}(A, B) := \mathcal{P}_{2\text{PI}}[\text{WickCov}_G(A, B)]. \quad (\text{A.56})$$

The first two coefficients of the exact interaction remainder in the convention of (4.10) are then

$$\Phi_1[m, G] = \mathbb{E}_G[\tilde{V}_1(m, \phi)], \quad (\text{A.57})$$

$$\Phi_2[m, G] = \mathbb{E}_G[\tilde{V}_2(m, \phi)] - \frac{1}{2} \text{Cov}_G^{2\text{PI}}(\tilde{V}_1(m, \phi), \tilde{V}_1(m, \phi)), \quad (\text{A.58})$$

and hence

$$\Sigma_1[m, G] = -2\nabla_G \Phi_1[m, G], \quad \Sigma_2[m, G] = -2\nabla_G \Phi_2[m, G]. \quad (\text{A.59})$$

All Gaussian trigonometric moments in these expressions follow from the multivariate Gaussian characteristic function. In particular, whenever the cubic Taylor tensors of V_1 are nonzero, the two-cubic-vertex contribution reduces to the mechanism in Proposition 5.1.

Proposition A.3 (Coefficient equivalence of the direct series and perturbative 2PI stationary branch)

Let $(m_\star(\varepsilon), G_\star(\varepsilon))$ be the exact moments of (A.51), and let $(m_{2\text{PI}}^{[p]}(\varepsilon), G_{2\text{PI}}^{[p]}(\varepsilon))$ be the stationary branch obtained from the complete perturbative 2PI functional (5.12) through order p , under the nondegeneracy assumptions of Theorem 5.2. Then the coefficients of $(m_{2\text{PI}}^{[p]}, G_{2\text{PI}}^{[p]})$ through ε^p are exactly the coefficients

obtained by truncating the direct moment expansions (A.65)–(A.66). In particular, the coefficients of the plotted order-2 dressed series coincide through $O(\varepsilon^2)$ with those of the stationary branch generated by the complete order-two perturbative 2PI functional. Equivalently, the plotted curve is the second-order Taylor polynomial of that 2PI stationary branch, not the finite- ε self-consistent solution of the truncated functional.

Proof The bounded perturbation $x - u = a_\varepsilon \sin u$ preserves an open linear/quadratic source neighbourhood, as shown in Section 7; hence the exact double Legendre transform exists locally in the stated sense for this family. Its physical stationary point reproduces the exact connected source derivatives $(m_\star, G_\star)$. Equations (A.65)–(A.66) are the Taylor coefficients of those exact moments. Lemma 5.2 and Corollary 5.1 imply that the complete perturbative 2PI stationary branch has the same Taylor coefficients through order p . Uniqueness of the local branch identifies the two coefficient sets. $\square$

Let b, c, S be defined in (7.13)–(7.14), and define

$$(C_1)_{ij} := \mathcal{G}_{ij}(c_i + c_j), \quad (\text{A.60})$$

$$(C_2)_{ij} := \Xi_{ij} - b_i b_j. \quad (\text{A.61})$$

Then the exact connected moments are

$$m_\star(\varepsilon) = \mu + a_\varepsilon b, \quad (\text{A.62})$$

$$G_\star(\varepsilon) = \mathcal{G} + a_\varepsilon C_1 + a_\varepsilon^2 C_2. \quad (\text{A.63})$$

For any analytic function $f(\varepsilon)$, let $[f]_{\leq p}$ denote its Taylor polynomial about $\varepsilon = 0$ through degree p . Then

$$a_\varepsilon^{[p]} := \sum_{r=1}^p \frac{(-1)^{r-1}}{r!} \varepsilon^r, \quad b_\varepsilon^{[p]} := [(a_\varepsilon)^2]_{\leq p}. \quad (\text{A.64})$$

The complete direct order- p moment series used in Figure 4 is therefore

$$m^{[p]}(\varepsilon) = \mu + a_\varepsilon^{[p]} b, \quad (\text{A.65})$$

$$G^{[p]}(\varepsilon) = \mathcal{G} + a_\varepsilon^{[p]} C_1 + b_\varepsilon^{[p]} C_2. \quad (\text{A.66})$$

For example,

$$a_\varepsilon^{[2]} = \varepsilon - \frac{1}{2} \varepsilon^2, \quad b_\varepsilon^{[2]} = \varepsilon^2. \quad (\text{A.67})$$

These are analytic Taylor coefficients of the exact physical moments, not quantities fitted from the plotted errors.

The exact physical self-energy displayed in Figures 1–3 is computed in the physical x coordinates from

$$\Sigma_\star = H_y[m_\star] - G_\star^{-1}, \quad (\text{A.68})$$

with $H_y[m_\star]$ obtained by differentiating (A.51). This makes the coordinate dependence of Remark 5.2 explicit: in the underlying Gaussian u coordinates the inverse connected two-point function is block banded, whereas the nonlinear physical x coordinates generate the off-band tail.